

Efficient Estimation in Semiparametric Accelerated Gap-Time Models for Recurrent Event Data

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Abstract

The accelerated failure time (AFT) model relates covariates to log-transformed event times under right censoring and extends naturally to recurrent events through the accelerated gap-time (AGT) formulation, providing a meaningful alternative to the Cox model. In many applications, such as reliability engineering and biomedical research, interventions between successive events may alter the timing of subsequent events, motivating models that explicitly accommodate such effects. We consider a broad class of semiparametric accelerated gap-time models for recurrent event data incorporating an effective age process, allowing a wide range of intervention mechanisms to be represented within a unified framework. To overcome estimation challenges arising from an infinite-dimensional baseline hazard and non-monotone score functions, we construct a sample-based weighted efficient score via parametric submodels. The resulting estimators are consistent and asymptotically normal. The proposed method is illustrated through simulation studies and an application to a biomedical dataset.

Keywords: semiparametric models; accelerated gap-time model; efficient estimation; survival analysis; recurrent event data; incomplete follow-up

1 Introduction

In reliability engineering and biomedical research, time to event data are routinely collected for units or subjects together with relevant covariates. In reliability studies, factors such as material properties, operating conditions, maintenance history, and usage intensity influence component lifetimes, while in biomedical studies variables including age, sex, blood pressure, cholesterol levels, and lifestyle habits are often associated with survival outcomes. Regression models are therefore widely used to quantify these relationships. The Cox [1] model is particularly popular because it links covariates multiplicatively to the hazard function.

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However, the proportional hazards assumption limits its ability to represent settings in which covariates accelerate or decelerate event times. The accelerated failure time model instead assumes that covariates act multiplicatively on the event time scale, which makes it an attractive alternative [2].

The AFT model written in log linear form is parametric when the error distribution is specified and semi-parametric (sAFT) when it is left unspecified. In the single event setting, Prentice [3] proposed estimation of the regression parameters using weighted linear rank estimating equations, and Tsiatis [4] established the large sample properties of estimators in this class. Buckley and James [5] introduced a least squares approach accommodating right censoring, and Ritov [6] studied the resulting estimating equations, showing that they yield consistent and asymptotically normal estimators and can attain the same efficiency as Prentice type linear rank procedures.

Recurrent event data, in which a subject may experience multiple events over time, are common in longitudinal studies. Strawderman [7] introduced a semiparametric accelerated gap-time (sAGT) model that relates the logarithm of inter event gap times to covariates through a linear structure, thereby describing how covariates stretch or compress the time between successive events. Estimation and inference were developed using semiparametric efficiency theory through projection onto the tangent space associated with the baseline hazard, extending Prentice type weighted linear rank methods to recurrent gap times. Consistency and asymptotic normality were established under Gehan [8] and log-rank [9] weights using empirical process and weak convergence arguments.

In many applications, interventions between successive events may change the timing of subsequent ones. In reliability engineering, for example, corrective actions such as part replacement, recalibration, or operational adjustments typically follow a failure. Effective interventions may lengthen the next gap time, whereas ineffective interventions may shorten it. A natural way to represent these effects is through an effective age concept that modifies the system’s age after intervention and thereby alters the evolution of future gap times.

The idea of effective age was introduced by Kijima et al. [10] and has been used in a variety of settings, including real estate [11–13], actuarial science [14], and medicine [15–17]. In this work, we develop a broad class of semiparametric accelerated gap time-models for recurrent event data that incorporates an effective age process, allowing a wide range of intervention mechanisms to be represented within a unified framework. This formulation includes the standard sAFT and sAGT models as special cases, corresponding to the *As Bad As Old* (ABAO) and *As Good As New* (AGAN) assumptions, respectively and it provides a flexible approach for analyzing recurrent event data under potentially imperfect interventions through appropriate specifications of the effective age process.

The rest of the paper is organized as follows. Section 2 presents the general class of semiparametric accelerated gap-time models and its formulation using counting processes. In Section 3, we extend this formulation to incorporate both calendar time and effective age time scales. Section 4 presents the proposed efficient estimation method. Section 5 illustrates the inference procedure using existing effective age processes from the literature, while Section 6 establishes the corresponding asymptotic properties. Section 7 provides results from simulation studies that evaluate the finite sample performance of the estimators. Section 8 illustrates the application of our model to a real world dataset from rehospitalization studies. Finally,

Section 9 concludes with a discussion of the numerical findings and potential directions for future research. All proofs of the theoretical developments are provided in the Appendix.

2 Postulated Model

All random quantities are defined on a common probability space $(\Omega, \mathcal{F}, P)$. We consider a study where the aging of n repairable systems is of interest, with the possibility of imperfect interventions after each failure or malfunction. Each system is observed over the interval $[0, \tau_i]$. Let $S_0 \equiv 0 < S_{i,1} < S_{i,2} < S_{i,3} < \dots$ denote the calendar event times of occurrences, and let the associated gap times be $T_{i,j} = S_{i,j} - S_{i,j-1}$. Since the duration of each intervention is considered negligible and is not included in the analysis, we assume that the calendar times $S_{i,j}$ represent the times at which interventions occur, while $T_{i,j}$ represents the time increment between the $(j-1)$ -th and j -th interventions. The τ_i 's are i.i.d. *end-of-monitoring times*, representing the point at which observation for each system ends, with unknown probability distribution function $G(\cdot)$. Since the observation mechanism in most studies is typically unknown, we assume that the τ_i 's are non-informative, ensuring that the survivor function of the end-of-monitoring times does not carry additional information about the distribution of system ages. It is also valuable to explore factors that may influence the timing of interventions. Each system i is associated with a p -dimensional vector of time-invariant covariates, $\mathbf{Z}_i = (Z_{i,1}, Z_{i,2}, \dots, Z_{i,p})'$. These covariates capture characteristics such as lifestyle habits, frequency of use, weather exposure, road conditions, and operating hours, all of which are presumed to impact the system's degradation process.

Let $\tilde{\mathcal{E}}_{i,0}, \tilde{\mathcal{E}}_{i,1}, \tilde{\mathcal{E}}_{i,3}, \dots$ be a sequence of baseline effective ages, where each $\tilde{\mathcal{E}}_{i,j}$ represents the effective age of the i -th system before intervention at the j -th recurrence or calendar time $S_{i,j}$, with a continuous baseline cumulative distribution function $F_0(\cdot)$. Under accelerated conditions, conditional on the covariate vector $\mathbf{Z}_i$, the effective ages $\mathcal{E}_{i,0}, \mathcal{E}_{i,1}, \mathcal{E}_{i,2}, \dots$ of the i -th system are observed, enabling inference to be made about the unobserved baseline effective ages. This relationship is formalized through the proposed model

$$\mathcal{E}_{i,j} = \tilde{\mathcal{E}}_{i,j} e^{-\theta'_0 \mathbf{Z}_i}, \quad (2.1)$$

for each intervention node $j \geq 1$ of the i -th system. The covariate vector $\mathbf{Z}_i$ directly influences the observed effective age $\mathcal{E}_{i,j}$, by either "accelerating" or "decelerating" it via an exponentiated weighted linear combination of the covariates. In essence, the covariates can compress or stretch the age of the system after accounting for the effects of interventions.

As a result, the data is characterized by $\mathcal{O}_i(\tau_i) = (K_i, S_{i,1}, \dots, S_{i,K_i}, \tau_i, \mathcal{E}_{i,j}, \mathbf{Z}_i)$ for $i = 1, \dots, n$, with $N_i^\dagger(s^-) = N_i((s \wedge \tau_i)^-)$. A counting process formulation of the problem leads to the following sequence of observable processes at time $s \in \mathcal{S}$: $\{(N_i^\dagger(s), Y_i^\dagger(s), \mathcal{E}_i(s), \mathbf{Z}_i), i = 1, \dots, n\}$, where $\{Y_i^\dagger(s) = I(\tau_i \geq s) : s \in \mathcal{S}\}$ denotes the "at-risk" process, indicating whether the system is still at risk at calendar time s , and $\{\mathcal{E}_i(s) : s \in \mathcal{S}\}$ represents the effective age process, reflecting the system's age under accelerated consideration. Hence, for each i -th system at calendar time s , we observe the number of interventions performed, the "at-risk" status, the system's age under accelerated conditions and the time-invariant factors that influence the system's aging process. The counterpart to the latter, without accelerated consideration,

is given by $\{\tilde{\mathcal{E}}_i(s) : s \in \mathcal{S}\}$, which is related to the effective ages under acceleration by the relation $\tilde{\mathcal{E}}_i(s) = \mathcal{E}_i(s)e^{\theta'_i \mathbf{Z}_i}$. Finally, let $\mathcal{F}_{s-} = \sigma\left\{N_i^\dagger(v), Y_i^\dagger(v), \mathcal{E}_i(v), \mathbf{Z}_i, i = 1, \dots, n; 0 \leq v < s\right\}$ for $s \in \mathcal{S}$, represent the full history of the observed processes up to, but not including, the calendar time s . The symbol $\sigma\{\cdot\}$ denotes the sigma-algebra generated by the processes and variables listed within the brackets, capturing the information available to the researchers just prior to time s . Additionally, the process $\mathbf{F} = \{\mathcal{F}_s, s \in \mathcal{S}\}$ is increasing and right-continuous, due to the right-continuity of the processes $N_i^\dagger(v)$, $Y_i^\dagger(v)$, and $\mathcal{E}_i(v)$.

We now present the first result of this work, Lemma 2.1, which specifies a probabilistic model for analyzing repairable systems data within a recurrent event data analysis framework.

Lemma 2.1. *For each $i = 1, \dots, n$, define the process*

$$M_i^\dagger(s) = N_i^\dagger(s) - A_i^\dagger(s, \boldsymbol{\theta}_0), \quad s \in \mathcal{S},$$

with

$$A_i^\dagger(s, \boldsymbol{\theta}) = \int_0^s \lambda_0(\tilde{\mathcal{E}}_i(v)) e^{\boldsymbol{\theta}' \mathbf{Z}_i} Y_i^\dagger(v) dv, \quad (2.2)$$

where $\lambda_0(\cdot)$ denoting an unknown continuous baseline hazard function associated with $F_0(\cdot)$ and $\boldsymbol{\theta}$ a p -dimensional regression parameter. Then, the processes $M_i^\dagger(s), i = 1, \dots, n$, are square-integrable, zero-mean martingales with respect to the filtration $\mathcal{F}_s$, and they are mutually orthogonal.

The proof of this lemma primarily relies on the Doob-Meyer Decomposition theorem and related martingale results, as detailed in [18] or [19]. The effective age framework provides a flexible and unifying approach for modeling systems subject to interventions. Semiparametric models based on effective age processes have been proposed [20, 21] for recurrent events under both multiplicative and additive intensity functions. However, their models do not incorporate covariate effects on the effective age within the $\mathbf{F}$ -compensator. This limitation is addressed in Lemma 2.1 under our proposed accelerated life model defined in (??). As for the form of the effective age process, it depends on the nature of the intervention. For example, *minimal intervention* ($\mathcal{E}(s) = s$) retains the system in an ABAO state, while *perfect intervention* ($\mathcal{E}(s) = s - S_{N^\dagger(s-)}$) resets it to an AGAN state. A more general formulation is given by Block et al. [22]: let $B_{i,1}, B_{i,2}, \dots$ be a sequence of i.i.d. Bernoulli(p) random variables, where a success indicates a perfect intervention and p may vary with time. Define the number of perfect interventions by $m_i(s) = \sum_{j=1}^{N_i^\dagger(s-)} B_{i,j}(s)$, and let the corresponding perfect intervention times be given by the random indices $0 \equiv \Omega_{i,0} < \Omega_{i,1} < \dots$, where $\Omega_{i,k} = \min\{j > \Omega_{i,k-1} : B_{i,j} = 1\}$. The effective age process is then defined as $\mathcal{E}_i(s) = s - S_{i,\Omega_{i,m_i(s-)-}}$. This construction recovers minimal intervention when $p = 0$ and perfect intervention when $p = 1$. The Brown-Proschan [23] model arises as a special case with constant success probability. Further generalizations include Kijima's Type I and Type II [10, 24] models, as well as the ARA models proposed by Doyen and Gaudoin [25], which offer greater flexibility for modeling imperfect intervention schemes. These frameworks have been further studied in [15, 26]

3 Doubly-Indexed Processes

Estimating the model parameters and establishing their asymptotic properties are challenging for two main reasons. First, the baseline hazard $\lambda_0(\cdot)$ depends on the effective age process $\tilde{\mathcal{E}}_i(\cdot)$. Second, semiparametric estimation must rely on counting and at-risk processes associated with the sequence of random effective age variables $\tilde{\mathcal{E}}_{i,j}$. To address these issues, we adopt the doubly-indexed processes introduced by Peña et al. [27], which facilitate transitioning between calendar time and effective age in gap time modeling. Over the domain $\mathcal{T} = [0, t^*]$, where t^* is a fixed maximum effective age, we define $R_i(s, t) = I\{\tilde{\mathcal{E}}_i(s) \leq t\}$, along with the associated processes:

$$N_i(s, t) = \int_0^s R_i(v, t) dN_i^\dagger(v), \quad A_i(s, t) = \int_0^s R_i(v, t) dA_i^\dagger(v, \boldsymbol{\theta}), \quad (3.1)$$

$$M_i(s, t) = \int_0^s R_i(v, t) dM_i^\dagger(v);$$

where, $A_i^\dagger(v, \boldsymbol{\theta})$ denotes the compensator of $N_i^\dagger(v)$, and $M_i^\dagger(v)$ is the associated martingale as defined in Lemma 2.1. Specifically, the doubly-indexed counting process $N_i(s, t)$ is a Lebesgue–Stieltjes integral, where $N_i^\dagger(s)$ experiences jumps at the observed event times $S_{i,1}, S_{i,2}, S_{i,3}, \dots$. Consequently, it takes the form

$$N_i(s, t \mid \boldsymbol{\theta}) = \sum_{j=1}^{K_i} I(\tilde{\mathcal{E}}_i(S_{i,j}) \leq t), \quad (3.2)$$

which counts the number of events observed over calendar time for which the corresponding effective age does not exceed t . Following this, Proposition 3.1 presents a change-of-variable technique that transforms stochastic integrals from calendar time to effective age, removing the dependence of the baseline hazard $\lambda_0(\cdot)$ on $\tilde{\mathcal{E}}_i(\cdot)$ in (2.2). To enable this, we restrict the effective age process between successive events for the i th system by defining $\mathcal{E}_{i,j}(s) = \mathcal{E}_i(s)I(S_{i,j-1} < s \leq S_{i,j})$ for s such that $Y_i^\dagger(s) > 0$. We assume that $\mathcal{E}_{i,j}(s)$ is invertible and differentiable on $(S_{i,j-1}, S_{i,j}]$, and define $I_{\mathcal{E}_{i,j}}(t; u, v) = I(\mathcal{E}_{i,j}(u^+) < t \leq \mathcal{E}_{i,j}(v))$ for any $u < v$. The remainder follows the approach of Peña et al. [27] and is omitted.

Proposition 3.1. *Let $h(\cdot)$ be a bounded integrable function. For $(s, t) \in \mathcal{S} \times \mathcal{T}$:*

$$\int_0^s h(\tilde{\mathcal{E}}_i(v)) \lambda_0(\tilde{\mathcal{E}}_i(v)) e^{\boldsymbol{\theta}' \mathbf{Z}_i} Y_i^\dagger(v) dv = \int_0^\infty h(r) \lambda_0(r) Y_i(s, r \mid \boldsymbol{\theta}) dr, \quad (3.3)$$

where for $i = 1, \dots, n$,

$$Y_i(s, r \mid \boldsymbol{\theta}) = \sum_{j=1}^{K_i} \frac{I_{\mathcal{E}_{i,j-1}}(re^{-\boldsymbol{\theta}' \mathbf{Z}_i}; S_{i,j-1}, S_{i,j})}{\mathcal{E}'_{i,j-1}(\mathcal{E}_{i,j-1}^{-1}(re^{-\boldsymbol{\theta}' \mathbf{Z}_i}))} + \frac{I_{\mathcal{E}_{i,K_i}}(re^{-\boldsymbol{\theta}' \mathbf{Z}_i}; S_{i,N_i^\dagger(s^-)}, s \wedge \tau_i)}{\mathcal{E}'_{i,K_i}(\mathcal{E}_{i,K_i}^{-1}(re^{-\boldsymbol{\theta}' \mathbf{Z}_i}))}. \quad (3.4)$$

Proposition 3.1 shows that $Y_i(s, t \mid \boldsymbol{\theta})$ is the generalized at-risk process corresponding to $N_i(s, t \mid \boldsymbol{\theta})$, and under the sAGT model of Strawderman [7], it coincides with the gap time version of $Y_i^\dagger(s)$, where

$\mathcal{E}_i(s) = s - S_{i, N_i^\dagger(s-)}$. The associated process $M_i(s, t \mid \boldsymbol{\theta})$ satisfies the Doob–Meyer decomposition $M_i(s, t \mid \boldsymbol{\theta}) = N_i(s, t \mid \boldsymbol{\theta}) - A_i(s, t \mid \boldsymbol{\theta})$, with $A_i(s, t \mid \boldsymbol{\theta}) = \int_0^t \lambda_0(w) Y_i(s, w \mid \boldsymbol{\theta}) dw$, where $Y_i(s, w \mid \boldsymbol{\theta})$ is defined in (3.4). Stochastic integration theory can be used to prove that $\{M_i(s, t \mid \boldsymbol{\theta}_0) : (s, t) \in \mathcal{S} \times \mathcal{T}\}$ is a square-integrable martingale for fixed t , with mean zero with respect to $\mathcal{F}_s$; however, $\{M_i(s, t \mid \boldsymbol{\theta}_0) : (s, t) \in \mathcal{S} \times \mathcal{T}\}$ is not a martingale for fixed s , due to the lack of a suitable filtration for the random effective age, though it remains a zero-mean process.

We now present Lemma 3.1 that allows us to perform time-scale transformations when integrating with respect to the doubly-indexed process $\{M_i(s, t \mid \boldsymbol{\theta}) : (s, t) \in \mathcal{S} \times \mathcal{T}\}$. The proof of Lemma 3.1 uses similar techniques as that of Proposition 3.1. It's therefore omitted.

Lemma 3.1. *Let $\mathbf{H}_i(s, t) = (H_{i1}(s, t), \dots, H_{ip}(s, t))'$ be vector-valued processes on $(\Omega, \mathcal{F}, \mathbf{P})$ with each $H_{ij}(\cdot, \cdot)$ being bounded, and $(s, t) \in \mathcal{S} \times \mathcal{T}$, for $i = 1, 2, \dots, n$. For each $t \in \mathcal{T}$, we assume that the process $\{\mathbf{H}_i(s, t) : s \in \mathcal{S}\}$ is $\mathbf{F}$ -predictable. Hence, we have:*

$$\int_0^s \mathbf{H}_i(s, \tilde{\mathcal{E}}_i(v)) M_i(dv, t) \equiv \int_0^t \mathbf{H}_i(s, r) M_i(s, dr \mid \boldsymbol{\theta}), \quad (3.5)$$

for $i = 1, 2, \dots, n$ and $(s, t) \in \mathcal{S} \times \mathcal{T}$. Moreover, Lemma 2.1 implies that:

$$\mathbb{E} \int_0^t \mathbf{H}_i(s, r) M_i(s, dr \mid \boldsymbol{\theta}_0) = 0, \quad (3.6)$$

$$\text{Var} \int_0^t \mathbf{H}_i(s, r) M_i(s, dr \mid \boldsymbol{\theta}_0) = \int_0^t \mathbb{E} \left[\mathbf{H}_i(s, r)^{\otimes 2} Y_i(r \mid \boldsymbol{\theta}_0) \right] \lambda_0(r) dr, \quad (3.7)$$

where $\mathbb{E}[\cdot]$ and $\text{Var}[\cdot]$ denote the expectation and variance operators, respectively.

To simplify the derivations in the estimation section, we introduce the following aggregated processes.

$$\begin{aligned} N(s, t \mid \boldsymbol{\theta}) &= \sum_{i=1}^n N_i(s, t \mid \boldsymbol{\theta}), & Y(s, t \mid \boldsymbol{\theta}) &= \sum_{i=1}^n Y_i(s, t \mid \boldsymbol{\theta}), \\ A(s, t \mid \boldsymbol{\theta}) &= \sum_{i=1}^n A_i(s, t \mid \boldsymbol{\theta}), & M(s, t \mid \boldsymbol{\theta}) &= \sum_{i=1}^n M_i(s, t \mid \boldsymbol{\theta}). \end{aligned}$$

4 Estimation Procedure

Censoring in failure time data restricts information on event times, complicating parameter estimation by introducing bias and reducing efficiency. The model involves a finite-dimensional regression parameter $\boldsymbol{\theta}_0$ and an infinite-dimensional baseline cumulative hazard function $\Lambda_0(\cdot)$, constituting a semiparametric framework. Treating Λ_0 as a nuisance parameter, we define the true parameter value as $\boldsymbol{\delta}_0 = (\boldsymbol{\theta}_0, \Lambda_0)$. Our method constructs an efficient score function for $\boldsymbol{\theta}_0$ by imposing orthogonality conditions, ensuring regular, asymptotically linear estimators and avoiding super-efficiency issues [28]. Additionally, we propose an Aalen-type estimator for $\Lambda_0(\cdot)$.

4.1 Maximum Likelihood Estimation

First, we construct the likelihood function based on the realizations of the counting process $N_i^\dagger(s)$, following the point-process approach developed by Jacod and others [19]. Specifically, the likelihood over the interval $[0, s]$ is

$$L(\boldsymbol{\delta}, s) \propto \prod_{v \in [0, s]} \prod_{i=1}^n \left\{ dA_i^\dagger(v, \boldsymbol{\theta}) \right\}^{dN_i^\dagger(v)} \exp \left\{ -A^\dagger(s, \boldsymbol{\theta}) \right\}, \quad (4.1)$$

where $dA_i^\dagger(v, \boldsymbol{\theta}) = \lambda_i(\mathcal{E}_i(v), \boldsymbol{\theta}) dv = Y_i^\dagger(v) \lambda_0(\tilde{\mathcal{E}}_i(v)) e^{\boldsymbol{\theta}' \mathbf{Z}_i} dv$, and $A^\dagger(s, \boldsymbol{\theta})$ denotes its aggregated version. $\prod$ is the product integral, a continuous analogue of $\prod$, also used to express the survivor function via the cumulative hazard [29]. Taking the log, differentiating with respect to $\boldsymbol{\theta}$, and assuming interchangeability of sum and integral, yields the score function for $\boldsymbol{\theta}_0$ at $\boldsymbol{\delta} = \boldsymbol{\delta}_0$:

$$U_{\boldsymbol{\theta}}^\psi(\boldsymbol{\delta}_0) = \sum_{i=1}^n \int_0^\infty \mathbf{Z}_i \psi(\tilde{\mathcal{E}}_i(v)) dM_i^\dagger(v), \quad (4.2)$$

as $s \rightarrow \infty$, where $\psi(r) = 1 + r\{\dot{\lambda}_0(r)/\lambda_0(r)\}$, with $\dot{\lambda}_0(u)$ the derivative of $\lambda_0(u)$ and $M_i^\dagger(v)$ as defined in Lemma 2.1.

Next, to derive the score function for the infinite-dimensional parameter $\Lambda_0(\cdot)$, we consider a family of parametric submodels $(\boldsymbol{\theta}_0, \Lambda_0) \rightarrow (\boldsymbol{\theta}_0, \Lambda_\gamma)$, with parametrization:

$$\Lambda_\gamma(t) = \int_0^t \lambda_0(w) e^{\gamma \kappa(w)} dw, \quad (4.3)$$

for $\gamma \in \mathbb{R}$ and $\kappa(\cdot)$ is some bounded element of a suitable Hilbert space $\mathbb{H}$ satisfying:

$$\mathbb{H} = \left\{ \kappa(\cdot) \in \mathbb{R} : \mathbb{E} \left[\int_0^\infty \kappa(r) dM_1(r | \boldsymbol{\theta}) \right] = 0, \int_0^\infty \kappa^2(r) \lambda_0(r) \mathbb{E} [Y_1(r | \boldsymbol{\theta})] dr < \infty \right\},$$

The true model $(\boldsymbol{\theta}_0, \Lambda_0)$ corresponds to $\gamma = 0$. Under parametrization (4.3), let $U_{\Lambda_0}^\kappa(\boldsymbol{\delta})$ denote the score function for $\Lambda_0(\cdot)$, obtained by differentiating the log-likelihood with respect to γ and evaluating at $\gamma = 0$. The resulting score function at the true parameter $\boldsymbol{\delta}_0$ is:

$$U_{\Lambda}^\kappa(\boldsymbol{\delta}_0) = \sum_{i=1}^n \int_0^\infty \kappa(\tilde{\mathcal{E}}_i(v)) dM_i^\dagger(v). \quad (4.4)$$

Hence, a parametric nuisance tangent space for $\Lambda_0(\cdot)$ is given by:

$$\mathcal{G}_\gamma = \left\{ \sum_{i=1}^n \int_0^\infty \mathcal{K}(\tilde{\mathcal{E}}_i(v)) dM_i^\dagger(v) : \mathcal{K}(\cdot) \in \mathbb{R}^p \text{ is an arbitrary function} \right\}.$$

with

$$\sum_{i=1}^n \int_0^\infty \mathcal{K}(\tilde{\mathcal{E}}_i(v)) dM_i^\dagger(v) = \sum_{i=1}^n \int_0^\infty \mathbf{B} \kappa(\tilde{\mathcal{E}}_i(v)) dM_i^\dagger(v) = \mathbf{B} U_{\Lambda}^\kappa(\boldsymbol{\delta}), \quad (4.5)$$

where $\mathbf{B} \in \mathbb{R}^p$ is arbitrary. Formally, the semiparametric nuisance tangent space $\mathcal{G}_{\text{sem}}$ is defined as the mean-square closure of the tangent spaces $\mathcal{G}_\gamma$ induced by all possible submodels. Intuitively, (4.5) indicates that $\mathcal{G}_{\text{sem}}$ can be characterized by allowing $\kappa(\cdot)$ to vary over the infinite-dimensional space of score functions $\mathbb{H}$ [30]. Note that $\mathcal{G}_{\text{sem}}$ is a subspace of $\mathbb{H}$.

We then restrict the log-likelihood, also used by Peña et al. [27] in a gap time setting, to effective ages $\tilde{\mathcal{E}}_i(s) \leq t$ over the intervals $[0, s]$ and $[0, t]$, using the indicator $R_i(s, t)$, motivated by technical considerations. Applying Lemmas 2.1 and 3.1, the resulting score functions for $\boldsymbol{\theta}_0$ and $\Lambda_0(\cdot)$ are expressed as integrals over the effective age.

$$U_{\boldsymbol{\theta}}^{\psi}(\boldsymbol{\delta}_0) = \sum_{i=1}^n \int_0^t \mathbf{Z}_i \psi(w) dM_i(w \mid \boldsymbol{\theta}_0) \quad (4.6)$$

and

$$U_{\Lambda}^{\kappa}(\boldsymbol{\delta}_0) = \sum_{i=1}^n \int_0^t \kappa(w) dM_i(w \mid \boldsymbol{\theta}_0). \quad (4.7)$$

Finally, we derive the efficient score function for $\boldsymbol{\theta}_0$ given by:

$$U_{\boldsymbol{\theta}}^{\psi, \text{eff}}(\boldsymbol{\delta}_0) = U_{\boldsymbol{\theta}}^{\psi}(\boldsymbol{\delta}_0) - U_{\Lambda}^{\xi}(\boldsymbol{\delta}_0),$$

where $U_{\Lambda}^{\xi}(\boldsymbol{\delta}_0)$ is the orthogonal projection of $U_{\boldsymbol{\theta}}^{\psi}(\boldsymbol{\delta}_0)$ onto the semiparametric nuisance tangent space for $\Lambda_0(\cdot)$. Specifically, $\xi(\cdot) = (\xi_1(\cdot), \dots, \xi_p(\cdot))'$ is the solution to the following orthogonality condition:

$$\mathbb{E} \left[\left\{ U_{\boldsymbol{\theta}}^{\psi}(\boldsymbol{\delta}_0) - U_{\Lambda}^{\xi}(\boldsymbol{\delta}_0) \right\} U_{\Lambda}^{\kappa'}(\boldsymbol{\delta}_0) \right] = \mathbf{0}_p, \text{ for any function } \kappa(\cdot) \in \mathbb{R}. \quad (4.8)$$

Let $\mathbf{z}^{(j)}(r \mid \boldsymbol{\theta}) = \mathbb{E} \left[\mathbf{Z}_1^{\otimes j} Y_1(r \mid \boldsymbol{\theta}) \right]$ for $j = 0, 1, 2$, where $\mathbf{x}^{\otimes k}$ denotes 1, a $p \times 1$ vector $\mathbf{x}$, and a matrix $\mathbf{x}\mathbf{x}'$ for $k = 0, 1, 2$, respectively. Proposition 4.1 then provides the efficient score function and the corresponding efficiency bound for estimating $\boldsymbol{\theta}_0$ —the semiparametric counterpart of the Cramér-Rao lower bound.

Proposition 4.1. *We define $\xi(r) = \psi(r)\bar{\mathbf{Z}}(r)$, where $\bar{\mathbf{Z}}(r) = \bar{\mathbf{Z}}(r \mid \boldsymbol{\theta}_0)$, and $\bar{\mathbf{Z}}(r \mid \boldsymbol{\theta}) = \mathbf{z}^{(1)}(r \mid \boldsymbol{\theta})/\mathbf{z}^{(0)}(r \mid \boldsymbol{\theta})$. Additionally, we assume $\int_0^\infty \psi^2(w)\lambda_0(w)dw < \infty$ to ensure integrability. Hence, the efficient score function for $\boldsymbol{\theta}_0$ at $\boldsymbol{\delta} = \boldsymbol{\delta}_0$ is given by:*

$$U_{\boldsymbol{\theta}}^{\psi, \text{eff}}(\boldsymbol{\delta}_0) = \sum_{i=1}^n \int_0^t \psi(w) \{ \mathbf{Z}_i - \bar{\mathbf{Z}}(w) \} dM_i(w \mid \boldsymbol{\theta}_0). \quad (4.9)$$

and its covariance matrix is:

$$\Sigma_{\boldsymbol{\theta}}^{\psi, \text{eff}} = \int_0^t \mathbb{E} \left[\{ \mathbf{Z}_1 - \bar{\mathbf{Z}}(w) \}^{\otimes 2} Y_1(w \mid \boldsymbol{\theta}_0) \right] \psi^2(w) \lambda_0(w) dw. \quad (4.10)$$

The semiparametric information bound $\mathcal{B}(\boldsymbol{\theta}_0)$ for estimating $\boldsymbol{\theta}_0$ is given by the inverse of $\Sigma_{\boldsymbol{\theta}}^{\psi, \text{eff}}$, provided it is positive definite.

An efficient score function and the corresponding semiparametric efficiency bound for $\Lambda_0(\cdot)$ can also be derived, following van der Vaart [30]. The following proposition summarizes these results.

Proposition 4.2. *Under the conditions of Proposition 4.1, and for $z^{(0)}(w | \theta_0) > 0$ for $w \in [0, t]$, the efficient score function for $\Lambda_0(\cdot)$ at $\delta = \delta_0$ is given by:*

$$U_{\Lambda}^{\kappa_{\text{eff}}}(\delta_0) = \sum_{i=1}^n \int_0^\infty \left[\frac{I\{w \leq t\}}{z^{(0)}(w | \theta_0)} - Q_{\psi}(w) \phi_i(w) \right] dM_i(w | \theta_0), \quad (4.11)$$

where $\phi_i(w) = Z_i - \bar{Z}(w)$, and $Q_{\psi}(w) = \Delta'_{\psi}(w) \mathcal{B}(\theta_0) \psi(w)$, with

$$\Delta_{\psi}(t) = \int_0^t \frac{z^{(1)}(w | \theta_0)}{z^{(0)}(w | \theta_0)} \psi(w) \lambda_0(w) dw. \quad (4.12)$$

and the efficiency bound $\mathcal{B}(\Lambda_0)$ for estimating $\Lambda_0(\cdot)$ at time t is the inverse of:

$$\int_0^t \frac{\lambda_0(w)}{z^{(0)}(w | \theta_0)} dw + \Delta'_{\psi}(t) \mathcal{B}(\theta_0) \Delta_{\psi}(t). \quad (4.13)$$

4.2 Estimation of δ_0

To estimate θ_0 using the efficient score function (4.9), we first replace $\bar{Z}(t | \theta)$ with its empirical estimate:

$$\bar{\mathbb{Z}}(t | \theta) = \frac{\mathbb{Z}^{(1)}(t | \theta)}{\mathbb{Z}^{(0)}(t | \theta)}, \text{ with } \mathbb{Z}^{(k)}(t | \theta) = \frac{1}{n} \sum_{i=1}^n Z_i^{\otimes k} Y_i(t | \theta), \quad k = 0, 1, 2. \quad (4.14)$$

As for $\psi(\cdot)$, it requires estimating the baseline hazard $\lambda_0(r) = f_0(r)/(1 - F_0(r))$ and its derivative $\dot{\lambda}_0(r)$. A common method, following Wand and Jones [31], applies kernel smoothing to a modified empirical distribution to avoid division by zero, with $\dot{\lambda}_0(r)$ obtained by differentiating the estimator [32]. This approach, however, is prone to bias, sensitive to bandwidth choice, and converges slowly. Kalbfleisch and Prentice [29] similarly highlighted challenges in estimating asymptotic variance due to the hazard derivative in AFT models. Alternatively, Ramlau-Hansen [33] proposed smoothing the Nelson-Aalen estimator, which Patil et al. [34] endorsed for its martingale form, practical value, and visual clarity. Still, Jin et al. [35] noted the limited reliability of nonparametric density estimation under censoring.

To address these challenges, Tsiatis [4] and Ritov [6] proposed an empirical weight function $\mathbb{W}(\cdot | \theta)$ as a substitute for the unknown $\psi(\cdot)$. By replacing $\psi(\cdot)$ with this function and using $\bar{\mathbb{Z}}(\cdot | \theta)$ in place of $\bar{Z}(\cdot | \theta)$ in (4.9), a sample-based weighted efficient score function is obtained:

$$U_{\theta}^{\mathbb{W}, \text{eff}}(\delta) = \sum_{i=1}^n \int_0^t \mathbb{W}(w | \theta) [Z_i - \bar{\mathbb{Z}}(w | \theta)] dN_i(w | \theta). \quad (4.15)$$

Letting $t \rightarrow \infty$, and using $N_i(s, t \mid \boldsymbol{\theta})$ expressed as a sum in (3.2), the score (4.15) becomes:

$$U_{\boldsymbol{\theta}}^{\mathbb{W}, \text{eff}}(\boldsymbol{\delta}) = \sum_{i=1}^n \sum_{j=1}^{K_i} \mathbb{W}(\tilde{\mathcal{E}}_{i,j} \mid \boldsymbol{\theta}) \left[\mathbf{Z}_i - \frac{\sum_{l=1}^n \mathbf{Z}_l Y_l(\tilde{\mathcal{E}}_{i,j} \mid \boldsymbol{\theta})}{\sum_{l=1}^n Y_l(\tilde{\mathcal{E}}_{i,j} \mid \boldsymbol{\theta})} \right], \quad (4.16)$$

with:

$$Y_l(\tilde{\mathcal{E}}_{i,j} \mid \boldsymbol{\theta}) = \sum_{r=1}^{K_l} \frac{I_{\mathcal{E}_{l,r-1}}(\tilde{\mathcal{E}}_{i,j} e^{-\boldsymbol{\theta}' \mathbf{Z}_l}; S_{l,r-1}, S_{l,r})}{\mathcal{E}'_{l,r-1}(\mathcal{E}_{l,r-1}^{-1}(\tilde{\mathcal{E}}_{i,j} e^{-\boldsymbol{\theta}' \mathbf{Z}_l}))} + \frac{I_{\mathcal{E}_{l,K_l}}(\tilde{\mathcal{E}}_{i,j} e^{-\boldsymbol{\theta}' \mathbf{Z}_l}; S_{l,K_l}, \tau_l)}{\mathcal{E}'_{l,K_l}(\mathcal{E}_{l,K_l}^{-1}(\tilde{\mathcal{E}}_{i,j} e^{-\boldsymbol{\theta}' \mathbf{Z}_l}))}. \quad (4.17)$$

The score function in (4.16) is similar to the weighted rank estimating function for censored data derived by Kalbfleisch and Prentice [36]. Therefore, $\hat{\boldsymbol{\theta}}$ can be identified as the zero-crossing or minimizer of $\|U_{\boldsymbol{\theta}}^{\mathbb{W}, \text{eff}}(\boldsymbol{\delta})\|$, where $\|\cdot\|$ denotes the Euclidean norm, though non-monotonicity may lead to multiple solutions. Since a plug-in estimator cannot be directly obtained, numerical methods like grid search work well for small covariates, while simulated annealing [37] is more efficient for high-dimensional covariates.

A common weight choice is $\mathbb{W}(t \mid \boldsymbol{\theta}) \equiv \mathbb{Z}^{(0)}(t \mid \boldsymbol{\theta})$, which corresponds to the Gehan [8] weight. This choice offers the advantage of improved numerical stability and guarantees the consistent convergence of $\hat{\boldsymbol{\theta}}$. We refer to the corresponding score function as $U_{\boldsymbol{\theta}}^{\mathbb{G}, \text{eff}}(\boldsymbol{\delta})$. Taking the limit as $t \rightarrow \infty$, equation (4.15) simplifies to:

$$\begin{aligned} U_{\boldsymbol{\theta}}^{\mathbb{G}, \text{eff}}(\boldsymbol{\delta}) &= \sum_{i=1}^n \int_0^\infty \mathbb{Z}^{(0)}(w \mid \boldsymbol{\theta}) [\mathbf{Z}_i - \bar{\mathbb{Z}}(w \mid \boldsymbol{\theta})] dN_i(w \mid \boldsymbol{\theta}) \\ &= \frac{1}{n} \sum_{i=1}^n \sum_{j=1}^{K_i} \sum_{l=1}^n Y_l(\tilde{\mathcal{E}}_{i,j} \mid \boldsymbol{\theta}) (\mathbf{Z}_i - \mathbf{Z}_l), \end{aligned} \quad (4.18)$$

where $Y_l(\tilde{\mathcal{E}}_{i,j} \mid \boldsymbol{\theta})$ is the generalized at-risk process defined in (4.17). Adopting the notation $\tilde{\mathcal{E}}_{lr}^{-i}(\boldsymbol{\theta}) = \tilde{\mathcal{E}}_{lr}(\boldsymbol{\theta}) e^{-\boldsymbol{\theta}' \mathbf{Z}_i}$, where

$$\tilde{\mathcal{E}}_{lr}(\boldsymbol{\theta}) = \begin{cases} \mathcal{E}_{l,r-1}(S_{l,r}) e^{\boldsymbol{\theta}' \mathbf{Z}_l} & \text{for } r \leq K_l, \\ \mathcal{E}_{l,K_l}(\tau_l) e^{\boldsymbol{\theta}' \mathbf{Z}_l} & \text{for } r = K_l + 1. \end{cases},$$

and

$$\tilde{\mathcal{E}}_{lr-1}(\boldsymbol{\theta}) = \begin{cases} \mathcal{E}_{l,r-1}(S_{l,r-1}) e^{\boldsymbol{\theta}' \mathbf{Z}_l} & \text{for } r \leq K_l, \\ \mathcal{E}_{l,K_l}(S_{l,K_l}) e^{\boldsymbol{\theta}' \mathbf{Z}_l} & \text{for } r = K_l + 1 \end{cases},$$

(4.18) can then be further expressed as:

$$U_{\boldsymbol{\theta}}^{\mathbb{G}, \text{eff}}(\boldsymbol{\delta}) = \frac{1}{n} \sum_{i=1}^n \sum_{j=1}^{K_i} \sum_{l=1}^n \sum_{r=1}^{K_l+1} \frac{I(\tilde{\mathcal{E}}_{lr-1}(\boldsymbol{\theta}) \leq \tilde{\mathcal{E}}_{ij}(\boldsymbol{\theta}) \leq \tilde{\mathcal{E}}_{lr}(\boldsymbol{\theta}))}{\mathcal{E}'_{l,r-1}(\mathcal{E}_{l,r-1}^{-1}(\tilde{\mathcal{E}}_{ij}(\boldsymbol{\theta})))} (\mathbf{Z}_i - \mathbf{Z}_l). \quad (4.19)$$

Other options include the Mantel [9] weight, $\mathbb{W}(t \mid \boldsymbol{\theta}) \equiv 1$, yielding a log-rank-type function $U_{\boldsymbol{\theta}}^{\mathbb{L}, \text{eff}}(\boldsymbol{\delta})$, and the Fleming and Harrington [18] family $\mathbb{W}(t \mid \boldsymbol{\theta}) \equiv \hat{F}_0^\beta(t) \{1 - \hat{F}_0^\beta(t)\}^\eta$, where $\hat{F}_0^\beta(t)$ serves as an estimator for $\tilde{F}_0(\cdot)$. While the log-rank score is optimal when $\lambda_0(r)$ is specified under a Weibull distribution,

it may still lack monotonicity and behave poorly like other general weights. Using the log-rank weight and letting $t \rightarrow \infty$, equation (4.15) simplifies to:

$$\begin{aligned} U_{\boldsymbol{\theta}}^{\mathbb{L}, \text{eff}}(\boldsymbol{\delta}) &= \sum_{i=1}^n \int_0^{\infty} [\mathbf{Z}_i - \bar{\mathbf{Z}}(w | \boldsymbol{\theta})] dN_i(w | \boldsymbol{\theta}) \\ &= \sum_{i=1}^n \sum_{j=1}^{K_i} \sum_{l=1}^n \frac{Y_l(\tilde{\mathcal{E}}_{i,j} | \boldsymbol{\theta})}{\sum_{l=1}^n Y_l(\tilde{\mathcal{E}}_{i,j} | \boldsymbol{\theta})} (\mathbf{Z}_i - \mathbf{Z}_l). \end{aligned} \quad (4.20)$$

This can be rewritten as:

$$U_{\boldsymbol{\theta}}^{\mathbb{L}, \text{eff}}(\boldsymbol{\delta}) = \sum_{i=1}^n \sum_{j=1}^{K_i} \sum_{l=1}^n p_l(\tilde{\mathcal{E}}_{i,j}(\boldsymbol{\theta}) | \boldsymbol{\theta}) (\mathbf{Z}_i - \mathbf{Z}_l), \quad (4.21)$$

with:

$$p_l(t | \boldsymbol{\theta}) = \frac{Y_l(t | \boldsymbol{\theta})}{\sum_{l=1}^n Y_l(t | \boldsymbol{\theta})} = \frac{Y_l(t | \boldsymbol{\theta})}{Y(t | \boldsymbol{\theta}_0)}, \quad t \geq 0. \quad (4.22)$$

The function $p_l(t | \boldsymbol{\theta})$ parallels Andersen and Gill [38], with the exponential term weighting the effective age. To avoid definition problems, we require that $Y(t | \boldsymbol{\theta}_0) > 0$ via $J(t | \boldsymbol{\theta}_0)$.

Finally, given a consistent estimator $\hat{\boldsymbol{\theta}}$, the baseline cumulative baseline hazard $\Lambda_0(t)$ can be estimated via a method-of-moments approach, yielding a Nelson-Aalen-type estimator:

$$\hat{\Lambda}_0(t | \boldsymbol{\theta}_0) = \int_0^t \frac{J(w | \boldsymbol{\theta}_0)}{Y(w | \boldsymbol{\theta}_0)} dN(w | \boldsymbol{\theta}_0), \quad (4.23)$$

where $J(w | \boldsymbol{\theta}_0) = I\{Y(w | \boldsymbol{\theta}_0) > 0\}$ ensures the estimator is well-defined. This form naturally arises by considering the restricted score function in (4.7) and imposing

$$\int_0^t \kappa(w) \{dN(w | \boldsymbol{\theta}_0) - Y(w | \boldsymbol{\theta}_0) d\Lambda_0(w)\} = 0, \quad \forall \kappa \in \mathbb{H},$$

which implies $d\Lambda_0(w) = dN(w | \boldsymbol{\theta}_0)/Y(w | \boldsymbol{\theta}_0)$, leading directly to (4.23). Thus, (4.23) may also be viewed as a maximum likelihood estimator of $\Lambda_0(t)$. However, using $\hat{\boldsymbol{\theta}}$ in place of $\boldsymbol{\theta}_0$ can introduce additional variability, potentially increasing the variance of $\hat{\Lambda}_0(t | \hat{\boldsymbol{\theta}})$ and reducing its efficiency.

5 Some Special Cases

We illustrate the inference procedure using specific choices of the effective age process from the literature, including those underlying the sAFT and sAGT models. Additional semiparametric examples are reviewed in Beutner [39].

Example 5.1. When $\mathcal{E}_i(s) = s$ (minimal intervention), we have $\mathcal{E}_i^{-1}(s) = s$ and $\mathcal{E}_i'(s) = 1$ and the efficient

score function (4.19) becomes:

$$U_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta}) = \frac{1}{n} \sum_{i=1}^n \sum_{j=1}^{K_i} \sum_{l=1}^n (\mathbf{Z}_i - \mathbf{Z}_l) I(\log \tau_l - \log S_{i,j} \geq \boldsymbol{\theta}'(\mathbf{Z}_i - \mathbf{Z}_l)),$$

This is the gradient of the convex objective function:

$$L_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta}) = \frac{1}{n} \sum_{i=1}^n \sum_{j=1}^{K_i} \sum_{l=1}^n \{\log \tau_l - \log S_{i,j} - \boldsymbol{\theta}'(\mathbf{Z}_i - \mathbf{Z}_l)\}^+, \quad (5.1)$$

with $a^+ = \max(a, 0)$, similar to the sAFT model of Lin et al. [40]. However, their approach does not assume independence of the baseline gap times or a common conditional distribution given $\mathbf{Z}$. The minimizers of $L_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta})$ and $\|U_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta})\|$ are asymptotically equivalent [41]. Alternatively, using the log-rank weight results in

$$U_{\boldsymbol{\theta}}^{\mathbb{L},\text{eff}}(\boldsymbol{\delta}) = \sum_{i=1}^n \sum_{j=1}^{K_i} \sum_{l=1}^n \frac{I(\log \tau_l - \log S_{i,j} \geq \boldsymbol{\theta}'(\mathbf{Z}_i - \mathbf{Z}_l))}{\sum_{l=1}^n I(\log \tau_l - \log S_{i,j} \geq \boldsymbol{\theta}'(\mathbf{Z}_i - \mathbf{Z}_l))} (\mathbf{Z}_i - \mathbf{Z}_l), \quad (5.2)$$

and given $\hat{\boldsymbol{\theta}}$, we estimate $\Lambda_0(t)$ using the formula:

$$\hat{\Lambda}_0(t | \hat{\boldsymbol{\theta}}) = \sum_{i=1}^n \sum_{j=1}^{K_i} \frac{I(\log t - \log S_{i,j} \geq \hat{\boldsymbol{\theta}}' \mathbf{Z}_i)}{\sum_{l=1}^n I(\log \tau_l - \log S_{i,j} \geq \hat{\boldsymbol{\theta}}'(\mathbf{Z}_i - \mathbf{Z}_l))}.$$

Example 5.2. When $\mathcal{E}_i(s) = s - S_{i,N_i^\dagger(s-)}$ (perfect intervention), we have $\tilde{\mathcal{E}}_{lr-1}(\boldsymbol{\theta}) = 0$, and $\mathcal{E}'_i(s) = 1$. Hence, $U_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta})$ in (4.19) is fully specified as follows:

$$U_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta}) = \frac{1}{n} \sum_{i=1}^n \sum_{j=1}^{K_i} \sum_{l=1}^n \sum_{r=1}^{K_l+1} (\mathbf{Z}_i - \mathbf{Z}_l) I(\tilde{\mathcal{E}}_{ij}(\boldsymbol{\theta}) \leq \tilde{\mathcal{E}}_{lr}(\boldsymbol{\theta})),$$

with:

$$\tilde{\mathcal{E}}_{lr}(\boldsymbol{\theta}) = \begin{cases} T_{l,r} e^{\boldsymbol{\theta}' \mathbf{Z}_l} & \text{for } r \leq K_l, \\ (\tau_l - S_{l,K_l}) e^{\boldsymbol{\theta}' \mathbf{Z}_l} & \text{for } r = K_l + 1. \end{cases},$$

which is the gradient of the convex objective function:

$$L_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta}) = \frac{1}{n} \sum_{i=1}^n \sum_{j=1}^{K_i} \sum_{l=1}^n \sum_{r=1}^{K_l+1} \left\{ \log \tilde{\mathcal{E}}_{lr}(\boldsymbol{\theta}) - \log \tilde{\mathcal{E}}_{ij}(\boldsymbol{\theta}) \right\}^+, \quad (5.3)$$

similar to the sAGT model [7]. However, in both examples, a reparameterization $\boldsymbol{\theta} \rightarrow -\boldsymbol{\theta}$ is required [35].

Under the log-rank weight, $U_{\theta}^{\mathbb{L}, \text{eff}}(\delta)$ becomes:

$$U_{\theta}^{\mathbb{L}, \text{eff}}(\delta) = \sum_{i=1}^n \sum_{j=1}^{K_i} \sum_{l=1}^n \frac{\sum_{r=1}^{K_l+1} I\left(\tilde{\mathcal{E}}_{ij}(\theta) \leq \tilde{\mathcal{E}}_{lr}(\theta)\right)}{\sum_{l=1}^n \sum_{r=1}^{K_l+1} I\left(\tilde{\mathcal{E}}_{ij}(\theta) \leq \tilde{\mathcal{E}}_{lr}(\theta)\right)} (\mathbf{Z}_i - \mathbf{Z}_l). \quad (5.4)$$

with:

$$\tilde{\mathcal{E}}_{lr}(\theta) = \begin{cases} T_{l,r} e^{\theta' \mathbf{Z}_l} & \text{for } r \leq K_l, \\ (\tau_l - S_{l,K_l}) e^{\theta' \mathbf{Z}_l} & \text{for } r = K_l + 1. \end{cases}$$

Given $\hat{\theta}$, the following formula can be used to estimate $\Lambda_0(t)$:

$$\hat{\Lambda}_0(t | \hat{\theta}) = \sum_{i=1}^n \sum_{j=1}^{K_i} \frac{I\left(\tilde{\mathcal{E}}_{ij}(\hat{\theta}) \leq t\right)}{\sum_{l=1}^n \sum_{r=1}^{K_l+1} I\left(\tilde{\mathcal{E}}_{ij}(\hat{\theta}) \leq \tilde{\mathcal{E}}_{lr}(\hat{\theta})\right)}.$$

Example 5.3. To motivate estimation under an effective age process that allows exploring a wider range of intervention degrees (minimal, perfect, and imperfect), we consider $\mathcal{E}_i(s)$ defined as:

$$\mathcal{E}_i(s) = \left(s - S_{i, N_i^{\dagger}(s-)}\right) + \sum_{k=1}^{N_i^{\dagger}(s-)} \left(\prod_{r=k}^{N_i^{\dagger}(s-)} (1 - D_{i,r}) \right) T_{i,k}. \quad (5.5)$$

Adekpedjou and Stocker [20] have shown that (5.5) is a special case of a broader class of linear effective age processes expressed as:

$$\mathcal{E}_i(s) = A_{i, N_i^{\dagger}(s-)} + B_{i, N_i^{\dagger}(s-)} \left[s - S_{i, N_i^{\dagger}(s-)} \right],$$

by defining $A_{i,0} = 1$, $A_{i,j} = \sum_{k=1}^j \left(\prod_{r=k}^j (1 - D_{i,r}) \right) T_{i,k}$, for all $j \geq 1$ and $B_{i,j} = 1$, for all j . From this perspective, the ratio in (4.19) takes on a more elegant and meaningful form, given by:

$$I \left\{ A_{l,r-1} e^{\theta' \mathbf{Z}_l} < (A_{i,j-1} + T_{i,j}) e^{\theta' \mathbf{Z}_i} \leq \{A_{l,r-1} + T_{l,r}\} e^{\theta' \mathbf{Z}_l} \right\} \text{ for } r \leq K_l,$$

$$I \left\{ A_{l,K_l} e^{\theta' \mathbf{Z}_l} < (A_{i,j-1} + T_{i,j}) e^{\theta' \mathbf{Z}_i} \leq \{A_{l,K_l} + (\tau_l - S_{l,K_l})\} e^{\theta' \mathbf{Z}_l} \right\} \text{ for } r = K_l + 1,$$

For simplicity, we will denote this indicator function by: $I_{\tilde{\mathcal{E}}_{l,r-1}}(t)$ for $r = 1, 2, \dots, K_l + 1$. As a result, $U_{\theta}^{\mathbb{G}, \text{eff}}(\delta)$ becomes:

$$U_{\theta}^{\mathbb{G}, \text{eff}}(\delta) = \frac{1}{n} \sum_{i=1}^n \sum_{j=1}^{K_i} \sum_{l=1}^n \sum_{r=1}^{K_l+1} (\mathbf{Z}_i - \mathbf{Z}_l) I_{\tilde{\mathcal{E}}_{l,r-1}} \left(\tilde{\mathcal{E}}_{ij}(\theta) \right),$$

which turns out to be the gradient of the convex objective function:

$$L_{\theta}^{\mathbb{G},\text{eff}}(\delta) = \frac{1}{n} \sum_{i=1}^n \sum_{j=1}^{K_i} \sum_{l=1}^n \sum_{r=1}^{K_l+1} \left\{ \log \tilde{\mathcal{E}}_{lr}(\theta) - \log \tilde{\mathcal{E}}_{ij}(\theta) \right\} I_{\tilde{\mathcal{E}}_{l,r-1}} \left(\tilde{\mathcal{E}}_{ij}(\theta) \right), \quad (5.6)$$

with:

$$\tilde{\mathcal{E}}_{lr}(\theta) = \begin{cases} \{A_{l,r-1} + T_{l,r}\} e^{\theta' \mathbf{Z}_l} & \text{for } r \leq K_l, \\ \{A_{l,K_l} + (\tau_l - S_{l,K_l})\} e^{\theta' \mathbf{Z}_l} & \text{for } r = K_l + 1. \end{cases}$$

Given $\hat{\theta}_G^K$, the following formula will be used to calculate $\hat{\Lambda}_0(t \mid \hat{\theta}_G^K)$. This approach will be illustrated in our simulation studies for various values of t .

$$\hat{\Lambda}_0(t \mid \hat{\theta}_G^K) = \sum_{i=1}^n \sum_{j=1}^{K_i} \frac{I \left(\tilde{\mathcal{E}}_{ij}(\hat{\theta}_G^K) \leq t \right)}{\sum_{l=1}^n \sum_{r=1}^{K_l+1} I_{\tilde{\mathcal{E}}_{l,r-1}} \left(\tilde{\mathcal{E}}_{ij}(\hat{\theta}_G^K) \right)}.$$

Figure 1 illustrates the objective function derived from the effective age process under Kijima's Model II with the Gehan weight function, showing increasingly smooth convexity as the sample size increases.

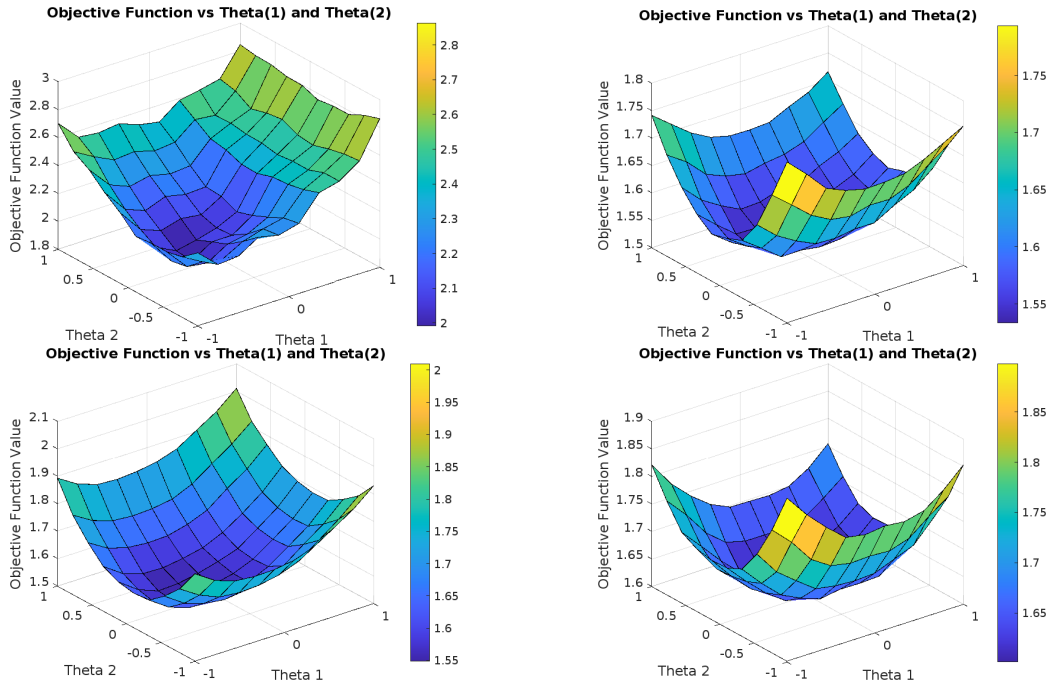


Figure 1: Gehan objective function under Kijima's Model II for varying sample sizes.

Examples 5.1, 5.2, and 5.3 illustrate that a closed-form solution for $\hat{\theta}$ is unattainable, regardless of the chosen weight function. Moreover, the score function $U_{\theta}^{\mathbb{L},\text{eff}}(\delta)$ is a weighted average of differences, where the presence of discontinuous indicator-based weights complicates the procedure. Also, while the Gehan

estimating function $U_{\theta}^{\mathbb{G},\text{eff}}(\delta)$ can be reformulated as a convex L_1 -loss minimization problem, ensuring the existence of a solution, this does not readily extend to the log-rank or generalized weighted log-rank score functions. Hence, by estimating θ using the Gehan weight, a consistent estimator can still be obtained for the log rank weight using linear programming techniques.

6 Asymptotic Results

In this section, we study the large sample properties [i.e $n \rightarrow \infty$] of the estimators $\hat{\theta}$ and $\hat{\Lambda}_0(\cdot | \hat{\theta})$ at time t .

6.1 Gehan-Weighted Estimation of θ_0

Theorem 6.1 summarizes the large sample behavior of $\hat{\theta}_G$ under Conditions C1-C7.

Theorem 6.1. *Let $\hat{\theta}_G = \arg \min_{\theta} \|U_{\theta}^{\mathbb{G},\text{eff}}(\delta)\|$ be the estimator of θ_0 . Under suitable regularity conditions, $\hat{\theta}_G$ exists with probability converging to one as $n \rightarrow \infty$ and satisfies $n^{1/2}(\hat{\theta}_G - \theta_0) \xrightarrow{d} N(0, \mathbb{A}_G)$, where $\mathbb{A}_G = \Phi_G^{-1} \Sigma_{\theta}^{\mathbb{G},\text{eff}} \Phi_G^{-1}$ with*

$$\Sigma_{\theta}^{\mathbb{G},\text{eff}} = \int_0^{\infty} \left[(z^{(0)}(w | \theta_0))^2 z^{(2)}(w | \theta_0) - z^{(0)}(w | \theta_0) (z^{(1)}(w | \theta_0))^{\otimes 2} \right] \lambda_0(w) dw,$$

and

$$\Phi_G = - \int_0^{\infty} \left[z^{(0)}(w | \theta_0) z^{(2)}(w | \theta_0) - (z^{(1)}(w | \theta_0))^{\otimes 2} \right] \psi(w) \lambda_0(w) dw. \quad (6.1)$$

To establish these results, we rely on Theorems 4.1 and 4.2 from Ritov [42]. Theorem 4.1 demonstrates the asymptotic normality of $\hat{\theta}_G$, though it requires an $n^{1/2}$ -consistent estimator, a condition that can be challenging to verify. Fortunately, the monotonicity of the Gehan-weighted score function $U_{\theta}^{\mathbb{G},\text{eff}}(\delta)$ (as the gradient of a convex function $L_{\theta}^{\mathbb{G},\text{eff}}(\delta)$) guarantees the existence of such an estimator and simplifies the proof of its $n^{1/2}$ -consistency. In a subsequent section, we will show how $\mathbb{A}_G$ can be consistently estimated using a numerical approximation method.

6.2 General Weighted Estimation of θ_0

The monotonicity of $U_{\theta}^{\mathbb{G},\text{eff}}(\delta)$ provides both computational and theoretical benefits that are lost with alternative weights like $\mathbb{W}(t | \theta) = 1$, despite its potential optimality. Constructing an estimator based on $\hat{\theta}_G$ may improve the estimation results. Unlike general weight functions that require θ_0 to lie in a compact set for consistency, the proposed estimator in Theorem 6.2 obtained via linear approximation does not, since $\hat{\theta}_G$ is already $n^{1/2}$ -consistent, thus simplifying computation. However, large-sample theory for general weights still demands stricter regularity conditions than those in Theorem 6.1. The one-step estimator's validity is ensured under Conditions C1-C10.

Theorem 6.2. *Let $\hat{\theta}_W$ be defined as $\hat{\theta}_W = \hat{\theta}_G - n^{-1} \hat{\Phi}_W^{-1} U_{\theta}^{\mathbb{W},\text{eff}}(\hat{\delta}_G)$, with $\hat{\Phi}_W$ a consistent estimator of Φ_W . Then $\hat{\theta}_W$ exists with probability converging to one as $n \rightarrow \infty$ and $n^{1/2}(\hat{\theta}_W - \theta_0) \xrightarrow{d} N(0, \mathbb{A}_W)$,*

with $\mathbb{A}_W = \Phi_W^{-1'} \Sigma_{\theta}^{\mathbb{W}, \text{eff}} \Phi_W^{-1}$, where

$$\begin{aligned}\Sigma_{\theta}^{\mathbb{W}, \text{eff}} &= \int_0^{\bar{\tau}} w^2(r \mid \theta_0) v(r \mid \theta_0) \mathbf{z}^{(0)}(r \mid \theta_0) \lambda_0(r) dr, \\ \Phi_W &= - \int_0^{\bar{\tau}} w(r \mid \theta_0) v(r \mid \theta_0) \mathbf{z}^{(0)}(r \mid \theta_0) \psi(r) \lambda_0(r) dr,\end{aligned}\tag{6.2}$$

with $v(r \mid \theta_0) = \mathbf{z}^{(2)}(r \mid \theta_0) / \mathbf{z}^{(0)}(r \mid \theta_0) - \{\mathbf{z}^{(1)}(r \mid \theta_0) / \mathbf{z}^{(0)}(r \mid \theta_0)\}^{\otimes 2}$; $\bar{\tau}$ and $w(r \mid \theta_0)$ defined in Conditions C8 and C9, respectively.

To compute $\hat{\theta}_W$, a consistent estimator of Φ_W is required. However, this does not incur additional computational cost, as an estimate of Φ_W is already necessary for computing $\mathbb{A}_W$. The procedures for estimating $\hat{\theta}_W$ and its variance are presented in the next section.

6.3 Estimating $\mathbb{A}_G$ and $\mathbb{A}_W$

The main challenge in estimating the variances in Theorems 6.1 and 6.2 is accurately estimating Φ_G and Φ_W due to their dependence on $\psi(\cdot)$. To address this issue, we adopt the variance estimation method proposed by Huang [43], which has been shown to yield reliable results. This technique, similar to numerical differentiation, is based on the asymptotic linearity of $U_{\theta}^{\mathbb{G}, \text{eff}}(\delta)$ in a neighborhood of θ_0 . Under Conditions C1–C5, $\hat{\Sigma}_{\theta}^{\mathbb{G}, \text{eff}}$ is consistent by Corollary A.1, and $\hat{\mathbb{A}}_G$ consistently estimates $\mathbb{A}_G$, as established in Theorem 6.1.

ALGORITHM 1

1. Compute $\hat{\theta}_G$ by minimizing $n^{-1} L_{\theta}^{\mathbb{G}, \text{eff}}(\delta)$, and estimate

$$\hat{\Sigma}_{\theta}^{\mathbb{G}, \text{eff}} = \int_0^{\infty} \left[\mathbb{Z}^{(0)}(w \mid \hat{\theta}_G) \mathbb{Z}^{(2)}(w \mid \hat{\theta}_G) - \left\{ \mathbb{Z}^{(1)}(w \mid \hat{\theta}_G) \right\}^{\otimes 2} \right] dN^{(0)}(w \mid \hat{\theta}_G).$$

This can be rewritten as $\hat{\Sigma}_{\theta}^{\mathbb{G}, \text{eff}} = \frac{1}{n} \sum_{i=1}^n \sum_{j=1}^{K_i} \varphi(\mathcal{E}_{i,j} e^{\hat{\theta}_G' \mathbf{Z}_i})$, with

$$\varphi(w) = \mathbb{Z}^{(0)}(w \mid \hat{\theta}_G) \mathbb{Z}^{(2)}(w \mid \hat{\theta}_G) - \left\{ \mathbb{Z}^{(1)}(w \mid \hat{\theta}_G) \right\}^{\otimes 2}.$$

2. Determine

$$\tilde{\theta}_G^k = \arg \min_{\theta} \left\| n^{-1} L_{\theta}^{\mathbb{G}, \text{eff}}(\delta) - h_k \right\|, \text{ for } k = 1, \dots, p,$$

where h_k is the k -th column of $\left(n^{-1} \hat{\Sigma}_{\theta}^{\mathbb{G}, \text{eff}} \right)^{1/2}$.

3. Calculate

$$\hat{\mathbb{A}}_G = \left(\hat{\mathbb{A}}_G^{1/2} \right)^2, \quad \text{with} \quad \hat{\mathbb{A}}_G^{1/2} = n^{1/2} \left(\tilde{\theta}_G^1 - \hat{\theta}_G, \dots, \tilde{\theta}_G^p - \hat{\theta}_G \right).$$

The main advantage of Algorithm 1 is that it eliminates the need to approximate Ψ_G . However, it involves minimizing a nonsmooth objective function. While this approach performs well with Gehan weights, extending it to general weight functions is challenging because the resulting estimating functions are non-monotone, complicating optimization and causing instability in the minimizers. To address this, Lyu et al. [44] proposed an induced smoothing technique that produces a smooth estimating function, allowing the application of standard numerical methods for estimating regression coefficients and variances in recurrent gap time data.

As for $\hat{\theta}_W$, it depends directly on the approximation of Φ_W . Hence, an alternative method that utilizes the asymptotic linearity of $U_{\theta}^{\mathbb{W},\text{eff}}(\delta)$ by sampling from the distribution of $\hat{\theta}_G$ can be used instead. Indeed, for any $0 < \Gamma < \infty$ and $\|\theta - \theta_0\| \leq \Gamma n^{-\frac{1}{2}}$, under the regularity conditions specified in Theorem 6.2, the following Taylor series expansion holds:

$$n^{-1}U_{\theta}^{\mathbb{W},\text{eff}}(\delta) - n^{-1}U_{\theta}^{\mathbb{W},\text{eff}}(\delta_0) = \Phi_W(\theta - \theta_0) + o_P\left(n^{-\frac{1}{2}}\right). \quad (6.3)$$

Thus, we randomly sample $\tilde{\theta}_k \sim N\left(\hat{\theta}_G, n^{-1}\hat{\mathbb{A}}_G\right)$ for $k = 1, \dots, B$ and applying equation (6.3) twice, we obtain:

$$n^{-1}U_{\theta}^{\mathbb{W},\text{eff}}(\tilde{\delta}_k) - n^{-1}U_{\theta}^{\mathbb{W},\text{eff}}(\hat{\delta}_G) = \Phi_W(\tilde{\theta}_k - \hat{\theta}_G) + \epsilon_k \quad (k = 1, \dots, B).$$

This implies that Φ_W can be viewed as the slope of a linear regression model with response $n^{-1}U_{\theta}^{\mathbb{W},\text{eff}}(\tilde{\delta}_k) - n^{-1}U_{\theta}^{\mathbb{W},\text{eff}}(\hat{\delta}_G)$ and regressors $\tilde{\theta}_k - \hat{\theta}_G$ for $k = 1, \dots, B$. Hence, Φ_W can be estimated as

$$\tilde{\Phi}_{W,B} = \left[(\mathfrak{P}'\mathfrak{P})^{-1} \mathfrak{P}'\Omega \right]^{(-1)},$$

where $A^{(-1)}$ represents matrix A without its first row and

$$\mathfrak{P} = \begin{bmatrix} \mathbf{1}_B & (\tilde{\Theta} - \hat{\theta}_G \mathbf{1}_B')' \end{bmatrix}, \quad \Omega = \left(n^{-1}\mathbf{U} - n^{-1}U_{\theta}^{\mathbb{W},\text{eff}}(\hat{\delta}_G)\mathbf{1}_B' \right),$$

where $\mathbf{1}_B$ is a $B \times 1$ vector of ones,

$$\tilde{\Theta} = [\tilde{\theta}_1, \tilde{\theta}_2, \dots, \tilde{\theta}_B], \text{ and } \mathbf{U} = [U_{\theta}^{\mathbb{W},\text{eff}}(\tilde{\delta}_1), \dots, U_{\theta}^{\mathbb{W},\text{eff}}(\tilde{\delta}_B)].$$

As a result, we have the following theorem, derived under Conditions C1–C10:

Theorem 6.3. *Let $\hat{\Phi}_W = \lim_{B \rightarrow \infty} \tilde{\Phi}_{W,B}$. Then $\hat{\Phi}_W \rightarrow \Phi_W$ in probability and $\hat{\mathbb{A}}_W = \hat{\Phi}_W^{-1'} \hat{\Sigma}_{\theta}^{\mathbb{W},\text{eff}} \hat{\Phi}_W^{-1}$ is a consistent estimator of $\mathbb{A}_W$, with $\hat{\Sigma}_{\theta}^{\mathbb{W},\text{eff}}$ given by*

$$\hat{\Sigma}_{\theta}^{\mathbb{W},\text{eff}} = \int_0^{\bar{\tau}} \mathbb{W}^2(w | \hat{\theta}_W) \left[\frac{\mathbb{Z}^{(2)}(w | \hat{\theta}_W)}{\mathbb{Z}^{(0)}(w | \hat{\theta}_W)} - \left\{ \frac{\mathbb{Z}^{(1)}(w | \hat{\theta}_W)}{\mathbb{Z}^{(0)}(w | \hat{\theta}_W)} \right\}^{\otimes 2} \right] \mathbb{N}^{(0)}(dw | \hat{\theta}_W).$$

Since $\tilde{\Phi}_{W,B}$ is not necessarily symmetric, we use the symmetric approximation

$$\hat{\Phi}_{W,B} = \frac{1}{2} \left(\tilde{\Phi}_{W,B} + \tilde{\Phi}_{W,B}' \right).$$

The computational algorithm for estimating $\hat{\theta}_W$ and $\hat{\mathbb{A}}_W$ is then summarized as follows.

ALGORITHM 2

1. Apply Algorithm 1 to determine $\hat{\theta}_G$ and obtain an estimate for $\hat{\mathbb{A}}_G$.
2. For a sufficiently large B , compute $\hat{\Phi}_{W,B}$, then estimate $\hat{\theta}_W$ using:

$$\hat{\theta}_W = \hat{\theta}_G - n^{-1} \hat{\Phi}_{W,B}^{-1} U_{\theta}^{\mathbb{W},\text{eff}} \left(\hat{\delta}_G \right).$$

3. Using $\hat{\Phi}_{W,B}$ as a substitute for $\hat{\Phi}_W$, compute $\hat{\mathbb{A}}_W$ following Theorem 6.3.

The primary drawback of this algorithm is the increased computational cost required to evaluate $n^{-1} U_{\theta}^{\mathbb{W},\text{eff}}(\tilde{\delta}_k)$ for each $k = 1, \dots, B$. For comparison, a modified version of Algorithm 1 using the log-rank weight will be used in the simulation to approximate $\mathbb{A}_{\text{LR}}$.

6.4 Estimation of $\Lambda_0(\cdot)$

Assuming $\hat{\theta}_W$ is obtained via Theorem 6.2, $\Lambda_0(t)$ is estimated by $\hat{\Lambda}_0(t \mid \hat{\theta}_W)$. In contrast, when using $\hat{\theta}_G$ derived from Theorem 6.1, use $\hat{\Lambda}_0(t \mid \hat{\theta}_G)$ instead and Conditions C9–C10 are no longer required.

Theorem 6.4. *Given $\hat{\theta}_W$, the process $n^{\frac{1}{2}} \left\{ \hat{\Lambda}_0(\cdot \mid \hat{\theta}_W) - \Lambda_0(\cdot) \right\}$ converges in distribution to a Gaussian process on $[0, \bar{\tau}]$ with the covariance function*

$$\mathbb{B}_{\Lambda_0}(r, t) = \int_0^{r \wedge t} \frac{\lambda_0(w)}{\mathbf{z}^{(0)}(w \mid \theta_0)} dw + \Delta'_{\psi}(r) \mathbb{A}_W \Delta_{\psi}(t), \quad (6.4)$$

where $\Delta_{\psi}(t)$ is defined in (4.12). Furthermore, for any finite $\Gamma > 0$, we have

$$\sup_{\substack{t \in [0, \bar{\tau}] \\ n^{\frac{1}{2}} \|\theta - \theta_0\| \leq \Gamma}} n^{\frac{1}{2}} \left| \hat{\Lambda}_0(t \mid \theta) - \hat{\Lambda}_0(t \mid \theta_0) + \Delta'_{\psi}(t)(\theta - \theta_0) \right| \xrightarrow{P} 0. \quad (6.5)$$

Setting $r = t$, a consistent variance estimator $\hat{\mathbb{B}}_{\Lambda_0}(t)$ is obtained using $\hat{\mathbb{A}}_W$ from Theorem 6.3 and a consistent estimator $\hat{\Delta}(t)$ for $\Delta_{\psi}(t)$.

The linear approximation in equation (6.5) suggests that $\Delta_{\psi}(t)$ can be estimated in a manner analogous to Theorem 6.3. Following Andersen et al. [19], a confidence interval $100(1 - \alpha)\%$ for $\Lambda_0(t)$ can then be obtained using the following:

$$\widehat{\Lambda}_0(t | \widehat{\boldsymbol{\theta}}_W) \exp \left[\pm \frac{z_{\alpha/2} \widehat{\mathbb{B}}_{\Lambda_0}(t)}{n^{1/2} \widehat{\Lambda}_0(t | \widehat{\boldsymbol{\theta}}_W)} \right]$$

6.5 Goodness-of-Fit Analysis

To assess the adequacy of the proposed recurrent event model, we compare regression parameter estimators obtained under two different weighting schemes, denoted by W_1 and W_2 . Let $\widehat{\boldsymbol{\theta}}_{W_1}$ and $\widehat{\boldsymbol{\theta}}_{W_2}$ denote the corresponding estimators. Under correct model specification, both estimators target the same underlying parameter vector $\boldsymbol{\theta}$, and therefore their difference should be small. Substantial discrepancies between the estimators may indicate potential model misspecification.

This idea can be viewed as a Hausman-type specification test [45], which evaluates whether two estimators that are consistent under the null hypothesis diverge when model assumptions are violated. Although Hausman-type tests are widely used in econometrics, similar ideas have also appeared in survival analysis. For example, Wei et al. [46] and Lin [47] considered related diagnostics for single-event Cox and AFT models, while Lin et al. [40] extended these ideas to recurrent event AFT models.

Under standard regularity conditions, if the proposed model holds, then the difference between the two estimators satisfies

$$n^{1/2}(\widehat{\boldsymbol{\theta}}_{W_1} - \widehat{\boldsymbol{\theta}}_{W_2}) \xrightarrow{d} N(\mathbf{0}, \boldsymbol{\Omega}),$$

where $\boldsymbol{\Omega}$ denotes the asymptotic covariance matrix of the difference. This result follows from Theorem 3.1 of Lin [47]. In our setting, $\boldsymbol{\Omega}$ can be consistently approximated by

$$\boldsymbol{\Omega} = \mathbb{A}_{W_1} - \mathbb{A}_{W_2},$$

where $\mathbb{A}_W$ denotes the asymptotic covariance matrix of the estimator obtained using weight function W .

In general, $\mathbb{A}_W$ takes the form

$$\mathbb{A}_W = \Phi_W^{-1'} \Sigma_{\boldsymbol{\theta}}^{\mathbb{W}, \text{eff}} \Phi_W^{-1},$$

with W representing either weighting scheme W_1 or W_2 .

Based on this result, we construct the following chi-square type statistic for assessing model adequacy:

$$S = (\widehat{\boldsymbol{\theta}}_{W_1} - \widehat{\boldsymbol{\theta}}_{W_2})' \boldsymbol{\Omega}^{-1} (\widehat{\boldsymbol{\theta}}_{W_1} - \widehat{\boldsymbol{\theta}}_{W_2}).$$

The covariance matrix $\boldsymbol{\Omega}$ must be positive definite. Under the null hypothesis of correct model specification, the statistic S is asymptotically central chi-square distributed with p degrees of freedom. Large values of S indicate substantial disagreement between the estimators obtained under W_1 and W_2 , and therefore provide evidence against model adequacy. This Hausman-type diagnostic offers a practical and computationally efficient goodness-of-fit assessment for complex recurrent event models, where conventional residual-based diagnostics may be difficult to apply.

7 Simulation Studies

We assess the performance of the estimators $\hat{\theta}_G$ and $\hat{\theta}_{LR}$ under the weight functions of Gehan and log-rank. The goal is to investigate the robustness of our proposed estimators in sAGT models. Although these models have previously been studied, they represent a special case of our proposed class of models. Their inclusion enables us to validate the performance of our general estimation procedure in well-understood settings. To that end, we simulate recurrent event data for n independent systems, each characterized by a single covariate Z generated independently. Event times are generated through model (2.1). The effective age process $\mathcal{E}(s)$ is defined to reflect how prior events and interventions affect the system's age. Each system is administratively monitored until a time τ drawn independently from a $\text{Uni}(0, 3.5)$ distribution, yielding approximately 20% incomplete follow-up.

We repeat each experiment over 500 replications for sample sizes $n \in \{50, 100\}$. The estimators $\hat{\theta}_G$ and $\hat{A}_G$ were computed using Algorithm 1, while Algorithm 2 was employed to obtain $\hat{\theta}_{LR}$ and $\hat{A}_{LR}$. The estimators are evaluated in terms of empirical bias, standard deviation, and the coverage probability of the 95% confidence intervals. The average estimated standard errors for $\hat{\theta}_G$ and $\hat{\theta}_{LR}$, obtained using Huang's approximation technique, are reported under the column $\mathbb{E}[\text{Est. SE}]$ (Huang). For the log-rank weight, the corresponding values under $\mathbb{E}[\text{Est. SE}]$ (Reg) refer to the approximations based on Theorem 6.3. These estimates were not computed for the Gehan weight. Estimation of the standard errors was based on $B_1 = 100$ simulated values for $\hat{\theta}_{LR}$ and $B_2 = 100$ for $\hat{\Lambda}(\cdot | \hat{\theta}_G)$ and $\hat{\Lambda}(\cdot | \hat{\theta}_{LR})$.

We consider the sAGT model, where the effective age resets after each event. Specifically, we define the effective age process as $\mathcal{E}_i(s) = s - S_{i, N_i(s-)}$, reflecting the assumption that each recurrence initiates a new life cycle, independent of past events. Event times are computed as cumulative sums of the gap times, beginning from zero. The gap times are generated according to $T_{i,j} \sim \tilde{T}_{i,j} e^{-0.8Z_i}$. The baseline hazard follows a Weibull distribution, resulting in a Power Law Process (PLP) for the initial intensity, with $\lambda_0(t) = \alpha\gamma t^{\gamma-1}$ and $\Lambda_0(t) = \alpha t^\gamma$, where $\alpha = \gamma = 1.5$. Evaluated at $t = 1$ and $t = 3$, we have $\Lambda_0(1) = 1.5$ and $\Lambda_0(3) \approx 7.79$. Covariates are generated as $Z_i \sim \mathcal{N}(0, 1)$, and each system is administratively monitored until a time τ , yielding approximately 20% incomplete follow-up. This simulation allows us to evaluate the performance of the estimation procedure when covariate effects act on the time between events rather than on calendar time. All simulations were conducted in MATLAB, with results summarized in Table 1.

For the regression parameter θ , both the log-rank and Gehan estimators exhibit stable finite-sample behavior across $n \in \{50, 100\}$ (Table 1). The absolute bias is moderate and changes little with n , with the log-rank estimator showing slightly smaller bias than the Gehan estimator. Empirical standard errors decrease as sample size increases, indicating improved precision for both weighting schemes. In contrast, the estimated standard errors are substantially larger than the empirical standard deviations, both under Huang's approximation and under the regression-based approximation available for the log-rank weight. This pronounced overestimation of variability leads to conservative inference: Wald-type confidence intervals for θ achieve perfect coverage for the log-rank estimator and slightly lower, yet still close-to-nominal, coverage for the Gehan estimator, while percentile bootstrap intervals remain essentially at the nominal 95% level for both methods. Figure 2 complements these numerical findings by showing approximately unimodal, near-symmetric distributions of $\hat{\theta}$ centered close to the true value $\theta = 0.8$, with visibly reduced dispersion at

$n = 100$. The bias boxplots in Figure 3 further confirm that the bias is predominantly negative and becomes more concentrated as the sample size increases.

For the cumulative baseline hazard, the results highlight a clear small-sample difficulty at early calendar times under the present incomplete follow-up design. At $t = 1$, both estimators substantially underestimate $\Lambda_0(1) = 1.5$, with absolute bias around 0.6 for both $n = 50$ and $n = 100$ (Table 1). Although the empirical standard error decreases with increasing sample size, the estimated standard errors remain larger, resulting in conservative Wald intervals with coverage above the nominal level, while percentile-bootstrap coverage remains stable near 95%. The lambda density plots in Figure 5 show strong overlap between weighting schemes at $t = 1$ and only a modest tightening from $n = 50$ to $n = 100$, consistent with the relatively unchanged bias and moderate reduction in empirical variability.

At the later time point $t = 3$, where the true value is $\Lambda_0(3) \approx 7.79$, the absolute bias is notably larger for $n = 50$ but decreases appreciably at $n = 100$ for both estimators (Table 1). Empirical variability is smaller at $t = 3$ than at $t = 1$ and decreases slightly with n , whereas the estimated standard errors are extremely conservative and increase with sample size, leading to perfect Wald coverage in all scenarios and near-nominal percentile coverage. This conservativeness is visible in Figures 4 and 5: while the boxplots show tighter interquartile ranges at $n = 100$ and medians moving closer to the target, the density panels at $t = 3$ display concentrated distributions whose centers remain below $\Lambda_0(3)$, reflecting persistent underestimation despite increased sample size.

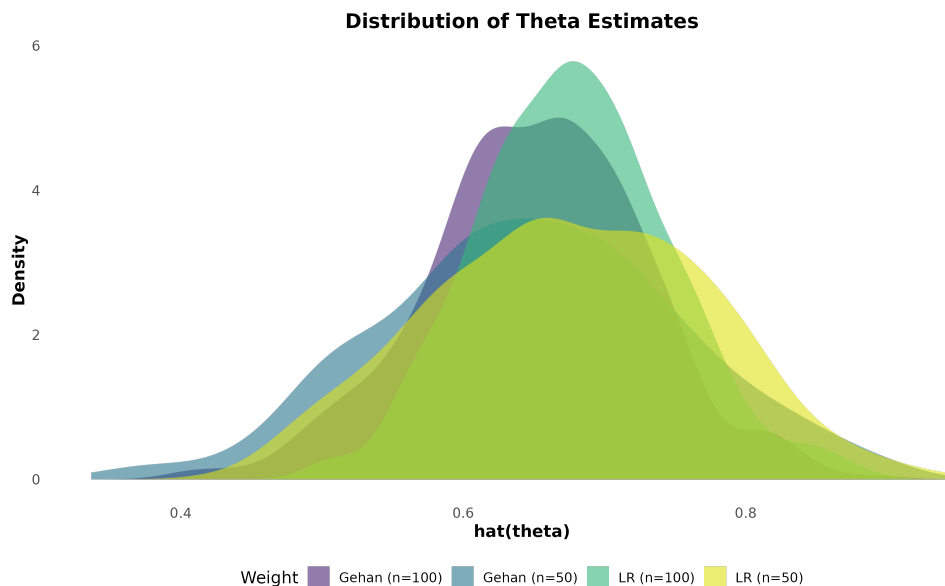


Figure 2: Distribution of $\hat{\theta}$ estimates under different weighting schemes and sample sizes. The plot shows the density of estimates for Gehan and Log-rank estimators.

Overall, the sAGT–PLP simulations demonstrate strong performance of both the Gehan- and log-rank-weighted procedures. Both approaches yield quite accurate point estimation for θ , with variability decreasing steadily as the sample size increases, reflecting the expected consistency of the estimators. Variance

Table 1: Simulation results for the sAGT-PLP model comparing Log-rank and Gehan weighting schemes at $n = 50$ and $n = 100$.

Metric	sAGT-PLP model			
	Log-rank		Gehan	
	$n = 50$	$n = 100$	$n = 50$	$n = 100$
Regression parameter θ				
$ \text{Bias}(\hat{\theta}) $	0.126	0.122	0.152	0.149
$\text{SE}(\hat{\theta})$	0.098	0.068	0.106	0.078
$\mathbb{E}\{\widehat{\text{SE}}(\hat{\theta})\}$ (Huang)	0.732	0.756	0.819	0.806
$\mathbb{E}\{\widehat{\text{SE}}(\hat{\theta})\}$ (Reg)	1.079	1.071	–	–
Coverage (Wald 95% CI)	1.000	1.000	0.972	0.944
Coverage (Percentile 95% CI)	0.952	0.952	0.952	0.952
Cumulative baseline hazard $\Lambda_0(t)$				
<i>At $t = 1$</i>				
$ \text{Bias}\{\hat{\Lambda}_0(1)\} $	0.607	0.597	0.661	0.654
$\text{SE}\{\hat{\Lambda}_0(1)\}$	0.255	0.182	0.277	0.207
$\mathbb{E}\{\widehat{\text{SE}}\{\hat{\Lambda}_0(1)\}\}$	2.436	2.442	2.067	2.052
Coverage (Wald 95% CI)	0.990	0.990	0.966	0.946
Coverage (Percentile 95% CI)	0.952	0.952	0.952	0.952
<i>At $t = 3$</i>				
$ \text{Bias}\{\hat{\Lambda}_0(3)\} $	1.743	1.038	1.743	1.036
$\text{SE}\{\hat{\Lambda}_0(3)\}$	0.145	0.121	0.145	0.111
$\mathbb{E}\{\widehat{\text{SE}}\{\hat{\Lambda}_0(3)\}\}$	9.054	12.798	9.036	12.794
Coverage (Wald 95% CI)	1.000	1.000	1.000	1.000
Coverage (Percentile 95% CI)	0.952	0.952	0.952	0.952

Notes: Results are based on 500 replications. $\text{SE}(\cdot)$ is the empirical standard error across replications, and $\widehat{\text{SE}}(\cdot)$ is a standard-error estimator computed within each replication. “Coverage” is the proportion of replications in which the nominal 95% confidence interval contains the true value.

estimation for both θ and $\Lambda_0(t)$ remains stable across settings, supporting reliable uncertainty quantification. Consequently, Wald-type intervals achieve coverage levels that are generally close to the nominal target, while percentile bootstrap intervals perform consistently well across parameters, weighting schemes, and sample sizes. The graphical summaries in Figures 2, 3, 4, and 5 further confirm these trends, showing a clear concentration of the estimators near the true values and progressively reduced dispersion as n increases.

8 Real Data Analysis

Colorectal cancer, a common malignancy in both men and women, typically originates from polyps in the colon or rectum, with risk factors including age, genetics, lifestyle, and conditions such as inflammatory bowel disease. Treatment generally begins with surgery, followed by chemotherapy or radiation; however, postoperative readmissions are frequent due to complications, recurrence, or treatment side effects. In this section, we apply our model to a dataset on hospital readmissions in colorectal cancer patients following

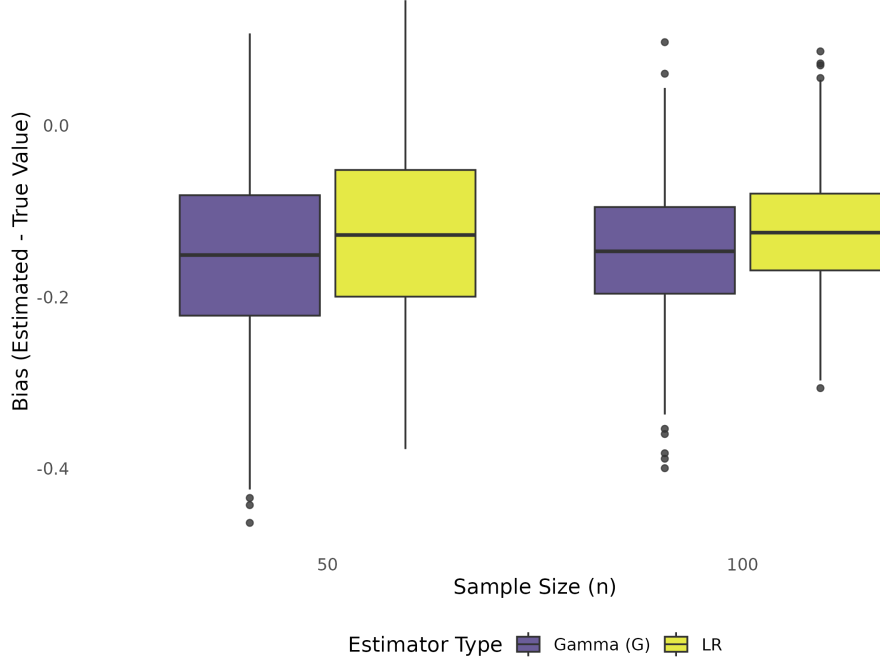


Figure 3: Boxplots of bias for $\hat{\theta}$ under Gehan and Log-rank weighting schemes across sample sizes.

surgery. The dataset includes 403 patients, divided into two groups based on chemotherapy status. The data come from a study by [48], published in the *Journal of Epidemiology and Community Health* and available in the R package `frailtypack`, which examines sex differences in readmission rates. We fit the proposed accelerated model in (2.1) to capture how patients age over time, incorporating treatment history and comorbid conditions, enabling a clearer understanding of readmission risk and supporting more informed clinical decisions. To evaluate the effectiveness of chemotherapy, we assess patient condition following readmission using three time-fixed covariates: chemotherapy status, Dukes' stage, and sex. Chemotherapy is coded as a binary variable (chemo = 1 for treated, chemo = 0 for untreated). Dukes' stage is categorized as 1 for stages A–B, 2 for stage C, and 3 for stage D, with lower values indicating less advanced cancer. Sex is coded as 0 for male and 1 for female. As noted by Stocker and Adekpedjou [21], model identifiability relies on directly observing the effective age process over time, and its absence adds complexity to the modeling task. For illustration, we consider the effective age process, always perfect interventions, corresponding to the sAGT model, with time-invariant covariates.

Table 2 and 3 present the parameter estimates, standard errors, and Wald-based confidence intervals. The estimates using the Log-rank weight are computed through Algorithm 2, employing $B_1 = 1500$ resampling iterations.

The results of the sAGT model applied to colorectal cancer readmission data are summarized in Tables 2 and 3. Table 2 presents the estimated regression coefficients, standard errors, and 95% Wald confidence intervals for three covariates, Chemo, Dukes, and Sex, under Gehan and Log-rank weighting schemes.

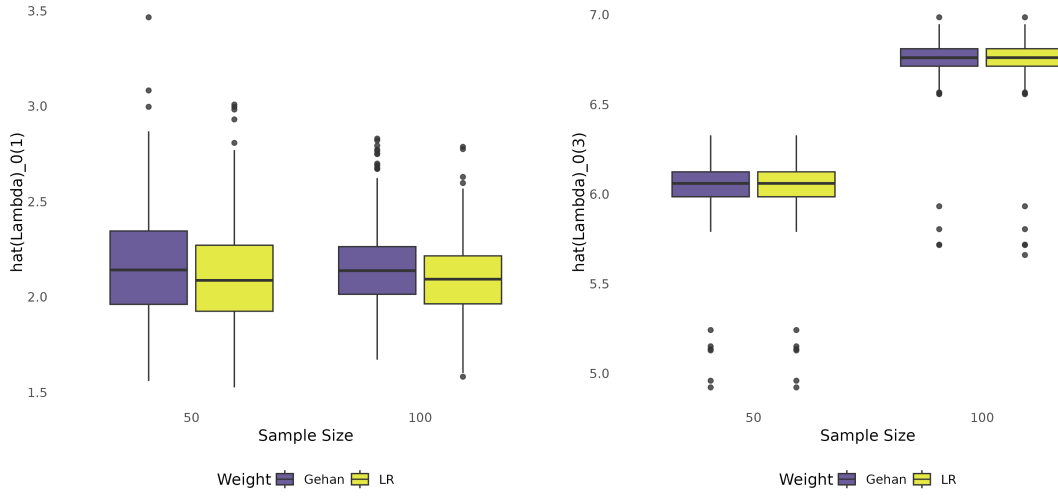


Figure 4: Baseline cumulative hazard estimates at $t = 1$ and $t = 3$ for Gehan and Log-rank estimators. The estimates correspond to $\hat{\Lambda}_0(1)$ (left) and $\hat{\Lambda}_0(3)$ (right).

Table 2: θ_0 estimates, standard errors, and 95% confidence intervals for the sAGT model applied to colorectal cancer readmission data under different weighting schemes. Under log-rank weighting, Huang and Reg methods yield identical point estimates; differences arise solely from the variance estimator.

Weight	Covariate	$\hat{\theta}$	Huang		Reg	
			SE	95% CI, Wald	SE	95% CI, Wald
Gehan	Chemo	-0.1459	2.8223	(-5.386, 5.678)	-	-
	Sex	0.5984	2.8147	(-6.115, 4.918)	-	-
	Dukes	-0.9343	1.8485	(-2.689, 4.558)	-	-
Log-rank	Chemo	-0.2046	1.9537	(-3.625, 4.034)	1.4517	(-2.641, 3.050)
	Sex	0.3335	2.2417	(-4.727, 4.060)	1.9211	(-4.099, 3.432)
	Dukes	-0.6956	1.8707	(-2.971, 4.362)	1.3856	(-2.020, 3.411)

Under Log-rank weighting, the Huang and Reg methods produce identical point estimates; differences arise solely from the variance estimator.

The Chemo covariate shows estimates close to zero with overlapping confidence intervals across all methods, suggesting no strong effect of chemotherapy on readmission gap times after adjusting for Dukes stage and sex. Dukes stage exhibits negative point estimates, indicating a potential reduction in the gap time between readmissions, although the relatively wide confidence intervals reflect some uncertainty when controlling for Chemo and sex. Sex has slightly positive estimates with wide confidence intervals that include zero, suggesting no statistically significant effect after accounting for Chemo and Dukes stage. Overall, Gehan–Huang tends to produce wider confidence intervals, while the log-rank methods provide more stable estimates with narrower intervals, reflecting greater precision.

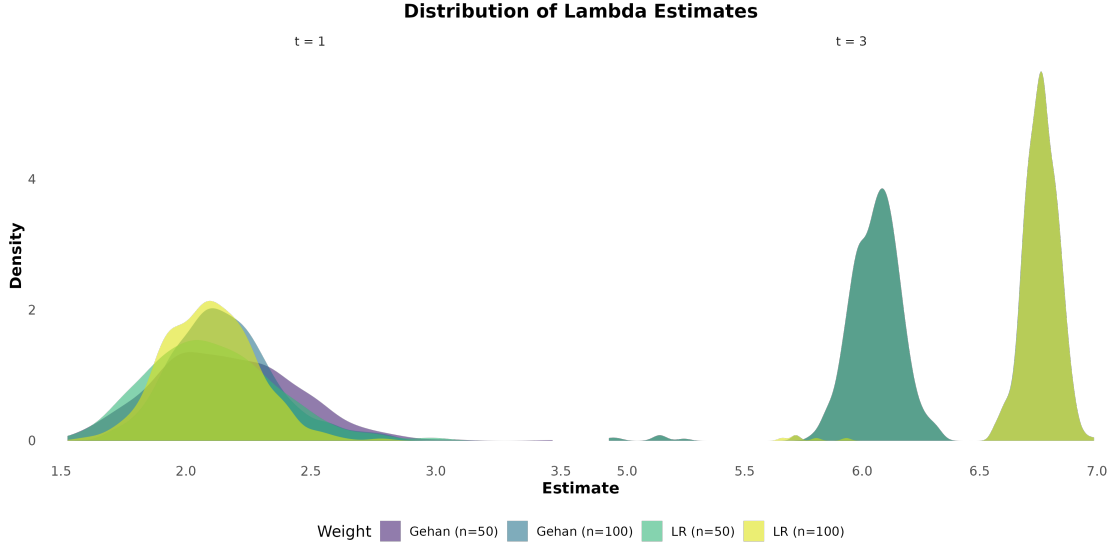


Figure 5: Density of baseline cumulative hazard estimates at $t = 1$ and $t = 3$ for Gehan and Log-rank estimators across sample sizes.

Table 3: Λ_0 estimates at three time points for the sAGT model applied to colorectal cancer readmission data, with standard errors and 95% confidence intervals.

Time	Gehan Weight			Log-rank Weight		
	$\hat{\Lambda}$	SE	95% CI, Wald	$\hat{\Lambda}$	SE	95% CI, Wald
$t_1 = 180$	0.2978	0.7950	(-1.260, 1.856)	0.3615	0.5908	(-0.797, 1.520)
$t_2 = 365$	0.4280	0.8642	(-1.266, 2.122)	0.4725	0.7503	(-0.998, 1.943)
$t_3 = 730$	0.5635	1.3855	(-2.152, 3.279)	0.6683	1.1535	(-1.593, 2.929)

Using the log-rank point estimates, the corresponding sAGT model can be expressed as

$$T_{i,j} = \tilde{T}_{i,j} \exp\{-0.205 \text{Chemo}_i + 0.333 \text{Sex}_i - 0.696 \text{Dukes}_i\}. \quad (8.1)$$

This formulation suggests that Dukes stage has the largest effect in magnitude and is associated with shorter gap times between readmissions, while chemotherapy also slightly reduces the gap time and sex is associated with a modest increase in the gap time. However, the confidence intervals containing zero indicate that these effects are not statistically significant.

Table 3 reports cumulative baseline hazard estimates at three time points for Gehan and Log-rank weights, along with standard errors and 95% confidence intervals. Since Wald intervals are reported on the original scale, they may extend below 0; alternative log-scale or transformation-based intervals would respect nonnegativity. Estimates increase over time, reflecting the accumulation of hazard, with small positive value at t_1 and larger values at t_2 and t_3 , consistent with the expected temporal pattern. Confidence intervals widen at later time points, particularly at t_3 , due to sparser risk sets at longer follow-up durations. Gehan weights exhibit slightly more variability than Log-rank weights, though both methods produce sim-

ilar point estimates, suggesting robustness of the cumulative hazard estimation. These results indicate that the Log-rank weighting tends to provide slightly more precise estimates than Gehan weighting, both for regression coefficients and cumulative baseline hazards, while the increasing uncertainty at later time points and for certain covariates is typical in survival analyses.

9 Discussion

To assess the adequacy of the sAGT model for the colorectal cancer readmission data, we compare the regression estimates obtained using the Gehan and log-rank weighting schemes. Let $\hat{\theta}_G$ denote the estimator obtained from the Gehan weight (Algorithm 1) and $\hat{\theta}_{LR}$ the estimator obtained from the log-rank weight (Algorithm 2). Under correct model specification, both estimators should target the same underlying parameter vector θ , so their difference is expected to be small, whereas substantial discrepancies could suggest model misspecification. As shown in Table 2, the estimated coefficients are broadly similar across the two weighting schemes for all three covariates, indicating that the results are relatively stable with respect to the choice of weight function. A formal Hausman-type comparison would require the covariance matrix of the difference $\Omega = \text{Var}(\hat{\theta}_G - \hat{\theta}_{LR})$. In practice, however, empirical estimates of Ω may fail to be positive definite in moderate samples due to the strong correlation between the estimators, which can lead to numerical instability when constructing quadratic-form test statistics. A possible remedy is to rely on the Moore–Penrose generalized inverse when the estimated covariance matrix is singular or ill-conditioned. In particular, one may replace the matrix inverse by its generalized inverse, thereby allowing the construction of test statistics and diagnostic quantities even in the presence of rank deficiency. This approach provides a stable alternative that preserves inferential validity without requiring ad hoc eigenvalue modifications, and may offer a more principled route for developing robust goodness-of-fit procedures for recurrent event models. Nevertheless, the close agreement between the Gehan and log-rank estimates observed in Table 2 provides empirical support for the adequacy of the sAGT model for these data.

Future work could also address the model’s non-smoothness by applying induced smoothing techniques [44], enabling smooth estimating functions for both Gehan and log-rank weights, reducing computational cost, and improving compatibility with standard numerical algorithms. Incorporating time-varying covariates, such as the Charlson comorbidity index, would provide a more dynamic view of patient health and better capture the evolving impact of treatment over time. To handle unobserved heterogeneity, a frailty component [49] could be integrated, supported by findings from [48], who reported lower readmission risks in women even after adjusting for clinical variables. Moreover, exploring nonlinear relationships using deep learning approaches, such as LSTM-Cox models [50], could improve the modeling of temporal dependencies in recurrent event data. A promising direction also lies in adapting the deep neural network-based AFT models proposed by [51], which use Gehan rank loss for single failure time data, to settings with incomplete follow-up in recurrent events, potentially enhancing both flexibility and predictive accuracy.

A Appendix

A.1 Regularity Conditions

The following regularity conditions ensure valid results and support asymptotic analysis. We assume that:

- C1. $f_0(\cdot)$ satisfying $F_0(\cdot) = \int_0^\cdot f_0(s)ds$ has continuous first and bounded second derivatives
- C2. The sequence of observable processes at time $s \in \mathcal{S}$: $\{(N_i^\dagger(s), Y_i^\dagger(s), \mathcal{E}_i(s), \mathbf{Z}_i), i = 1, \dots, n\}$ are independent and identically distributed.
- C3. $\tau_1 \mid \mathbf{Z}_1$ is a possibly degenerate distribution $G(\cdot \mid \mathbf{Z}_1)$.
- C4. The end of follow-up time is non-informative.
- C5. $\mathbf{Z}_1$ is almost surely bounded by a nonrandom constant Γ .
- C6. $\mathbf{z}^{(j)}(t \mid \boldsymbol{\theta})$, for $j = 0, 1, 2$, is continuous for $(t, \boldsymbol{\theta}) \in \mathbb{R}^+ \times \mathcal{N}(\boldsymbol{\theta}_0)$, where $\mathcal{N}(\boldsymbol{\theta}_0)$ is some neighbourhood of $\boldsymbol{\theta}_0$.
- C7. Φ_G in (6.1) is a nonsingular matrix.
- C8. $\bar{\tau}$ satisfies $\inf \{t \in [0, \bar{\tau}] : \mathbf{z}^{(0)}(t \mid \boldsymbol{\theta}_0)\} > 0$.
- C9. For $(t, \boldsymbol{\theta}) \in [0, \bar{\tau}] \times \mathcal{N}(\boldsymbol{\theta}_0)$, $\mathbb{W}(t \mid \boldsymbol{\theta})$ is of bounded variation and there exists a continuous, bounded deterministic function $w(t \mid \boldsymbol{\theta})$ such that $|\mathbb{W}(t \mid \boldsymbol{\theta}) - w(t \mid \boldsymbol{\theta})|$ converges to zero uniformly in probability.
- C10. Φ_W in (6.2) is a nonsingular.

A.2 Proofs of Section 4 Propositions

Proof of Proposition 4.1. The orthogonality condition (4.8) simplifies to:

$$\mathbb{E} \left[\left\{ U_{\boldsymbol{\theta}_k}^\psi(\boldsymbol{\delta}_0) - U_{\Lambda}^{\xi_k}(\boldsymbol{\delta}_0) \right\} U_{\Lambda}^{\kappa}(\boldsymbol{\delta}_0) \right] = 0, \text{ for } k = 1, \dots, p, \quad \xi_k(\cdot) \in \mathbb{H}, \quad (\text{A.1})$$

and any function $\kappa(\cdot) \in \mathbb{R}$ not identically zero. Without loss of generality, we consider $k = 1$, as the argument holds for all $k = 1, \dots, p$. To solve for $\xi_1(\cdot)$, we begin by rewriting the left-hand side of equation (A.1) using equations (4.2) and (4.4), which define $U_{\boldsymbol{\theta}_1}^\psi(\boldsymbol{\delta}_0)$ and $U_{\Lambda}^{\kappa}(\boldsymbol{\delta}_0)$, respectively. Since the $M_i^\dagger(\cdot)$ are orthogonal $\mathbf{F}$ -martingales by Lemma 2.1, and under Condition C2, applying Proposition 3.1 yields:

$$LHS \equiv n \int_0^t \mathbb{E} \left[\{Z_{11}\psi(w) - \xi_1(w)\} \kappa(w) \lambda_0(w) Y_1(w \mid \boldsymbol{\theta}_0) \right] dw.$$

It follows that

$$(\text{A.1}) \iff \psi(w) \mathbb{E}[Z_{11} Y_1(w \mid \boldsymbol{\theta}_0)] - \xi_1(w) \mathbb{E}[Y_1(w \mid \boldsymbol{\theta}_0)] = 0$$

$$\Longleftrightarrow \xi_1(w) = \psi(w) \bar{Z}_1(w),$$

with $\bar{Z}_1(w) = \mathbb{E}[Z_{11}Y_1(w | \boldsymbol{\theta}_0)] / \mathbb{E}[Y_1(w | \boldsymbol{\theta}_0)]$. The solution $\xi_1(\cdot)$ is an element of $\mathbb{H}$. In fact, by Condition C5, Z_1 is uniformly bounded. Furthermore, Condition A6 ensures that $z^{(0)}(r | \boldsymbol{\theta}_0) \neq 0$, which implies $\mathbb{E}[Y_1(r | \boldsymbol{\theta}_0)] = 0$ cannot be zero. Thus, using the score in (4.6) on the effective age scale, we obtain the efficient score function:

$$U_{\boldsymbol{\theta}}^{\psi, \text{eff}}(\boldsymbol{\delta}_0) = \sum_{i=1}^n \int_0^t \psi(w) \{Z_i - \bar{Z}(w)\} dM_i(w | \boldsymbol{\theta}_0).$$

Finally, the information bound $\mathcal{B}(\boldsymbol{\theta}_0)$ is the inverse of the efficient information matrix $\Sigma_{\boldsymbol{\theta}}^{\psi, \text{eff}}$, the covariance of $U_{\boldsymbol{\theta}}^{\psi, \text{eff}}(\boldsymbol{\delta}_0)$, computed via martingale properties:

$$\begin{aligned} \Sigma_{\boldsymbol{\theta}}^{\psi, \text{eff}} &= n^{-1} \mathbb{E} \left[U_{\boldsymbol{\theta}}^{\psi, \text{eff}}(\boldsymbol{\delta}_0) U_{\boldsymbol{\theta}}^{\psi, \text{eff}}(\boldsymbol{\delta}_0) \right] \\ &= \int_0^t \mathbb{E} \left[\{Z_1 - \bar{Z}(w)\}^{\otimes 2} Y_1(w | \boldsymbol{\theta}_0) \right] \psi^2(w) \lambda_0(w) dw, \end{aligned} \quad (\text{A.2})$$

which ends the proof. $\square$

Proof of Proposition 4.2. The efficient score function for $\Lambda_0(\cdot)$ is defined as:

$$U_{\Lambda}^{\kappa, \text{eff}}(\boldsymbol{\delta}_0) = U_{\Lambda}^{\nu}(\boldsymbol{\delta}_0) - \Pi U_{\Lambda}^{\nu}(\boldsymbol{\delta}_0), \quad (\text{A.3})$$

where $\Pi U_{\Lambda}^{\nu}(\boldsymbol{\delta}_0) = \mathbb{E} \left[U_{\Lambda}^{\nu}(\boldsymbol{\delta}_0) U_{\boldsymbol{\theta}}^{\psi, \text{eff}}(\boldsymbol{\delta}_0) \right] \left\{ \mathbb{E} \left[U_{\boldsymbol{\theta}}^{\psi, \text{eff}}(\boldsymbol{\delta}_0) U_{\boldsymbol{\theta}}^{\psi, \text{eff}}(\boldsymbol{\delta}_0) \right] \right\}^{-1} U_{\boldsymbol{\theta}}^{\psi, \text{eff}}(\boldsymbol{\delta}_0)$, is the projection of $U_{\Lambda}^{\nu}(\boldsymbol{\delta}_0)$ onto the space spanned by $U_{\boldsymbol{\theta}}^{\psi, \text{eff}}(\boldsymbol{\delta}_0)$. Substituting the formula for $\Pi U_{\Lambda}^{\nu}(\boldsymbol{\delta}_0)$ into (A.3) and using (A.2), we obtain:

$$U_{\Lambda}^{\kappa, \text{eff}}(\boldsymbol{\delta}_0) = U_{\Lambda}^{\nu}(\boldsymbol{\delta}_0) - \frac{1}{n} \mathbb{E} \left[U_{\Lambda}^{\nu}(\boldsymbol{\delta}_0) U_{\boldsymbol{\theta}}^{\psi, \text{eff}}(\boldsymbol{\delta}_0) \right] \mathcal{B}(\boldsymbol{\theta}_0) U_{\boldsymbol{\theta}}^{\psi, \text{eff}}(\boldsymbol{\delta}_0), \quad (\text{A.4})$$

where $U_{\Lambda}^{\nu}(\boldsymbol{\delta}_0) = \sum_{i=1}^n \int_0^{\infty} \frac{I\{\tilde{\mathcal{E}}_i(v) \leq t\}}{z^{(0)}\{\tilde{\mathcal{E}}_i(v) | \boldsymbol{\theta}_0\}} dM_i^{\dagger}(v)$, for $\nu(r) = I\{r \leq t\} / z^{(0)}(r | \boldsymbol{\theta}_0)$. Using Proposition 3.1, we show that $\mathbb{E} \left[U_{\Lambda}^{\nu}(\boldsymbol{\delta}_0) U_{\boldsymbol{\theta}}^{\psi, \text{eff}}(\boldsymbol{\delta}_0) \right] = n \Delta_{\psi}(t)$, where $\Delta_{\psi}(t)$ is given in (4.12). This follows from the orthogonality of the $M_i^{\dagger}(\cdot)$ by Lemma 2.1, Condition C2, which ensures i.i.d. integrands, and subsequent use of Proposition 3.1, making the left-hand side:

$$\begin{aligned} LHS &\equiv n \mathbb{E} \left[\int_0^{\infty} \left(\frac{I\{w \leq t\}}{z^{(0)}\{w | \boldsymbol{\theta}_0\}} \right) Z_1 \psi(w) \lambda_0(w) Y_1(w | \boldsymbol{\theta}_0) dw \right] \\ &= n \int_0^t \frac{z^{(1)}\{w | \boldsymbol{\theta}_0\}}{z^{(0)}\{w | \boldsymbol{\theta}_0\}} \psi(w) \lambda_0(w) dw. \end{aligned}$$

Note that $\Delta_{\psi}(t)$ and $U_{\boldsymbol{\theta}}^{\psi, \text{eff}}(\boldsymbol{\delta}_0)$ are p -column vectors while $\mathcal{B}(\boldsymbol{\theta}_0)$ is a $p \times p$ square matrix. Consequently,

$\Delta'_\psi(t)\mathcal{B}(\boldsymbol{\theta}_0)\psi(w) \{ \mathbf{Z}_i - \bar{\mathbf{Z}}(w) \}$ is a scalar and (A.4) is well-defined, simplifying to:

$$U_\Lambda^{\kappa_{\text{eff}}}(\boldsymbol{\delta}_0) = \sum_{i=1}^n \int_0^\infty \left[\frac{I\{w \leq t\}}{\mathbf{z}^{(0)}(w | \boldsymbol{\theta}_0)} - \Delta'_\psi(w)\mathcal{B}(\boldsymbol{\theta}_0)\psi(w) \{ \mathbf{Z}_i - \bar{\mathbf{Z}}(w) \} \right] dM_i(w | \boldsymbol{\theta}_0),$$

which gives (4.11). The information bound $\mathcal{B}(\Lambda_0)$ for estimating $\Lambda_0(t)$ requires computing $\text{Var} [n^{-1/2}U_\Lambda^{\kappa_{\text{eff}}}(\boldsymbol{\delta}_0)]$. Using the effective age representation of $U_\Lambda^{\kappa_{\text{eff}}}(\boldsymbol{\delta}_0)$, along with the variance formula in (3.7) and Condition C2, this variance is given by:

$$\int_0^\infty \mathbb{E} \left[\left(\frac{I\{w \leq t\}}{\mathbf{z}^{(0)}(w | \boldsymbol{\theta}_0)} - \Delta'_\psi(w)\mathcal{B}(\boldsymbol{\theta}_0)\psi(w) \{ \mathbf{Z}_1 - \bar{\mathbf{Z}}(w) \} \right)^2 \right] \lambda_0(w) Y_1(w | \boldsymbol{\theta}_0) dw.$$

Expanding and simplifying the squared term inside the expectation as follows:

$$\begin{aligned} \text{Var} [n^{-1/2}U_\Lambda^{\kappa_{\text{eff}}}(\boldsymbol{\delta}_0)] &= \int_0^t \frac{\mathbb{E}[Y_1(w | \boldsymbol{\theta}_0)]\lambda_0(w)}{\mathbf{z}^{(0)}(w | \boldsymbol{\theta}_0)^2} dw - 2 \times \int_0^t \Delta'_\psi(w)\mathcal{B}(\boldsymbol{\theta}_0)\psi(w) \\ &\quad \times \left\{ \frac{\mathbb{E}[\mathbf{Z}_1 Y_1(w | \boldsymbol{\theta}_0)]}{\mathbf{z}^{(0)}(w | \boldsymbol{\theta}_0)} - \bar{\mathbf{Z}}(w) \frac{\mathbb{E}[Y_1(w | \boldsymbol{\theta}_0)]}{\mathbf{z}^{(0)}(w | \boldsymbol{\theta}_0)} \right\} \lambda_0(w) dw \\ &\quad + \int_0^t \mathbb{E} \left[[\Delta'_\psi(w)\mathcal{B}(\boldsymbol{\theta}_0)\psi(w) \{ \mathbf{Z}_1 - \bar{\mathbf{Z}}(w) \}]^2 Y_1(w | \boldsymbol{\theta}_0) \right] \lambda_0(w) dw \\ &= \int_0^t \frac{\lambda_0(w)}{\mathbf{s}^{(0)}(w | \boldsymbol{\theta}_0)} dw + \Delta'_\psi(w)\mathcal{B}(\boldsymbol{\theta}_0)\Delta_\psi(w), \end{aligned}$$

yields the information bound $\mathcal{B}(\Lambda_0)$ in (4.13). □

A.3 Auxiliary Results of Section 6

We present several auxiliary results, considering all probabilities and expectations with respect to $\mathbb{P}_0$, the probability measure corresponding to $\boldsymbol{\delta}_0$. Let Θ be an open convex subset of $\mathbb{R}^p$. The proofs of Theorems A.1 and A.2 are built upon the empirical process theory established in [52].

Theorem A.1. *Let $\boldsymbol{\theta} \in \Theta$ be fixed, and for $t \in \mathbb{R}^+$, define*

$$\mathbb{N}^{(k)}(t | \boldsymbol{\theta}) = \frac{1}{n} \sum_{i=1}^n \mathbf{Z}_i^{\otimes k} N_i(t | \boldsymbol{\theta}) \quad (k = 0, 1, 2). \quad (\text{A.5})$$

When viewed as processes over $t \in \mathbb{R}^+$, both $\mathbb{N}^{(k)}(t | \boldsymbol{\theta})$ in (A.5) and

$$\mathbb{Z}^{(k)}(t | \boldsymbol{\theta}) = \frac{1}{n} \sum_{i=1}^n \mathbf{Z}_i^{\otimes k} Y_i(t | \boldsymbol{\theta}), \quad (k = 0, 1, 2), \quad (\text{A.6})$$

are $\mathbb{P}_0$ -Donsker.

Theorem A.2. *Both (A.5) and (A.6) are $\mathbb{P}_0$ -Donsker as processes indexed by $(t, \boldsymbol{\theta}) \in \mathbb{R}^+ \times \Theta$.*

The corollary below follows directly from Theorems A.1 and A.2.

Corollary A.1. *Under Conditions C1–C5, for $k = 0, 1, 2$, the following convergence in probability holds:*

$$\sup_{(t, \boldsymbol{\theta}) \in \mathbb{R}^+ \times \Theta} \left\| \mathbb{N}^{(k)}(t \mid \boldsymbol{\theta}) - \mathbf{n}^{(k)}(t \mid \boldsymbol{\theta}) \right\| \rightarrow 0, \quad \sup_{(t, \boldsymbol{\theta}) \in \mathbb{R}^+ \times \Theta} \left\| \mathbb{Z}^{(k)}(t \mid \boldsymbol{\theta}) - \mathbf{z}^{(k)}(t \mid \boldsymbol{\theta}) \right\| \rightarrow 0,$$

where $\mathbf{n}^{(k)}(t \mid \boldsymbol{\theta}) = \mathbb{E} \left[\mathbf{Z}_1^{\otimes k} N_1(t \mid \boldsymbol{\theta}) \right]$. These results remain valid if the supremum is taken only over $t \in \mathbb{R}^+$ while keeping $\boldsymbol{\theta} \in \Theta$ fixed.

We now give a useful approach for evaluating $\mathbf{n}^{(k)}(t \mid \boldsymbol{\theta})$ and $\mathbf{z}^{(k)}(t \mid \boldsymbol{\theta})$, for $k = 0, 1, 2$.

Lemma A.1. *Let $\boldsymbol{\theta} \in \mathbb{R}^p$ be arbitrary. Then, for each $i = 1, \dots, n$, the counting and at-risk processes satisfy the transformations:*

$$N_i(t \mid \boldsymbol{\theta}) = N_i \left(te^{(\boldsymbol{\theta}_0 - \boldsymbol{\theta})' \mathbf{Z}_i} \mid \boldsymbol{\theta}_0 \right), \quad Y_i(t \mid \boldsymbol{\theta}) = Y_i \left(te^{(\boldsymbol{\theta}_0 - \boldsymbol{\theta})' \mathbf{Z}_i} \mid \boldsymbol{\theta}_0 \right). \quad (\text{A.7})$$

Furthermore, for $k = 0, 1, 2$, the expectation of the weighted counting process can be rewritten as:

$$\mathbf{n}^{(k)}(t \mid \boldsymbol{\theta}) = \mathbb{E} \left[\mathbf{Z}_1^{\otimes k} \int_0^{te^{(\boldsymbol{\theta}_0 - \boldsymbol{\theta})' \mathbf{Z}_1}} Y_1(w \mid \boldsymbol{\theta}_0) \lambda_0(w) dw \right]. \quad (\text{A.8})$$

Proof. We prove the results in (A.7) step by step. Starting from (3.2), we have:

$$\begin{aligned} N_i(t \mid \boldsymbol{\theta}) &= \sum_{j=1}^{K_i} I(\tilde{\mathcal{E}}_i(S_{i,j}) \leq te^{(\boldsymbol{\theta}_0 - \boldsymbol{\theta})' \mathbf{Z}_i}) \\ &= N_i(te^{(\boldsymbol{\theta}_0 - \boldsymbol{\theta})' \mathbf{Z}_i} \mid \boldsymbol{\theta}_0). \end{aligned}$$

Next, from (3.4), as $s \rightarrow \infty$, we have:

$$\begin{aligned} Y_i(t \mid \boldsymbol{\theta}) &= \sum_{j=1}^{K_i} \frac{I_{\mathcal{E}_{i,j-1}}(te^{(\boldsymbol{\theta}_0 - \boldsymbol{\theta})' \mathbf{Z}_i} \cdot e^{-\boldsymbol{\theta}'_0 \mathbf{Z}_i}; S_{i,j-1}, S_{i,j})}{\mathcal{E}'_{i,j-1}(\mathcal{E}_{i,j-1}^{-1}(te^{(\boldsymbol{\theta}_0 - \boldsymbol{\theta})' \mathbf{Z}_i} \cdot e^{-\boldsymbol{\theta}'_0 \mathbf{Z}_i}))} + \frac{I_{\mathcal{E}_{i,K_i}}(te^{(\boldsymbol{\theta}_0 - \boldsymbol{\theta})' \mathbf{Z}_i} \cdot e^{-\boldsymbol{\theta}'_0 \mathbf{Z}_i}; S_{i,K_i}, \tau_i)}{\mathcal{E}'_{i,K_i}(\mathcal{E}_{i,K_i}^{-1}(te^{(\boldsymbol{\theta}_0 - \boldsymbol{\theta})' \mathbf{Z}_i} \cdot e^{-\boldsymbol{\theta}'_0 \mathbf{Z}_i}))} \\ &= Y_i(te^{(\boldsymbol{\theta}_0 - \boldsymbol{\theta})' \mathbf{Z}_i} \mid \boldsymbol{\theta}_0). \end{aligned}$$

Finally, to prove (A.8), note that the process $\{M_i(t \mid \boldsymbol{\theta}_0) : t \in \mathcal{T}\}$ is not a martingale but has zero mean. Therefore, for $k = 0, 1, 2$,

$$\begin{aligned} \mathbf{n}^{(k)}(t \mid \boldsymbol{\theta}) &= \mathbb{E} \left[\mathbf{Z}_1^{\otimes k} \left\{ M_1(te^{(\boldsymbol{\theta}_0 - \boldsymbol{\theta})' \mathbf{Z}_1} \mid \boldsymbol{\theta}_0) + A_1(te^{(\boldsymbol{\theta}_0 - \boldsymbol{\theta})' \mathbf{Z}_1} \mid \boldsymbol{\theta}_0) \right\} \right] \\ &= \mathbb{E} \left[\mathbf{Z}_1^{\otimes k} \int_0^{te^{(\boldsymbol{\theta}_0 - \boldsymbol{\theta})' \mathbf{Z}_1}} Y_1(w \mid \boldsymbol{\theta}_0) \lambda_0(w) dw \right], \end{aligned}$$

which ends the proof. □

A.4 Proofs of Section 6 Theorems

Proof of Theorem 6.1. The efficient score function for θ can be rewritten as:

$$U_{\theta}^{\mathbb{G},\text{eff}}(\delta) = \int_0^\infty \mathbb{Z}^{(0)}(w | \theta) d\mathbb{N}^{(1)}(w | \theta) - \int_0^\infty \mathbb{Z}^{(1)}(w | \theta) d\mathbb{N}^{(0)}(w | \theta).$$

First, by applying Theorem A.1 and Corollary A.1, we can use Lemma 1 from [53]. Consequently, there exists a function ϕ such that:

$$n^{-\frac{1}{2}} U_{\theta}^{\mathbb{G},\text{eff}}(\delta_0) = n^{-\frac{1}{2}} \sum_{i=1}^n \phi_i + o_p(1),$$

where $\phi_i = \int_0^\infty \{z^{(0)}(w | \theta_0) Z_i - z^{(1)}(w | \theta_0)\} dM_i(w | \theta_0)$, and satisfies the condition:

$$\mathbb{E}[\phi_1] = \mathbb{E} \left[\int_0^\infty \{z^{(0)}(w | \theta_0) Z_1 - z^{(1)}(w | \theta_0)\} dM_1(w | \theta_0) \right] = 0,$$

by (3.6). Using Lemma 3.1. Also, from (3.7), we have:

$$\begin{aligned} \text{Var}(\phi_1) &= \int_0^\infty \mathbb{E} \left[\left\{ z^{(0)}(w | \theta_0) Z_1 - z^{(1)}(w | \theta_0) \right\}^{\otimes 2} Y_1(w | \theta_0) \right] \lambda_0(w) dw \\ &= \int_0^\infty \left[\left\{ z^{(0)}(w | \theta_0) \right\}^2 z^{(2)}(w | \theta_0) - z^{(0)}(w | \theta_0) \left\{ z^{(1)}(w | \theta_0) \right\}^{\otimes 2} \lambda_0(w) \right] dw, \end{aligned}$$

which shows that $\text{Var}(\phi_1) = \Sigma_{\theta}^{\mathbb{G},\text{eff}}$. The central limit theorem implies that $n^{-\frac{1}{2}} U_{\theta}^{\mathbb{G},\text{eff}}(\delta_0) \xrightarrow{d} N(0, \Sigma_{\theta}^{\mathbb{G},\text{eff}})$, with Conditions C1 and C6 ensuring that $\|\Sigma_{\theta}^{\mathbb{G},\text{eff}}\| < \infty$. Next, under Conditions C1–C6, along with Corollary A.1 and Lemma A.1, the function $n^{-1} U_{\theta}^{\mathbb{G},\text{eff}}(\delta)$ converges uniformly in probability to:

$$\begin{aligned} \mathcal{U}_{\theta}^{\mathbb{G},\text{eff}}(\delta) &= \int_0^\infty \left\{ z^{(0)}(w | \theta) \mathbb{E}[Z_1 Y_1(w | \theta) \lambda_0\{w c_1(\theta)\} c_1(\theta)] \right. \\ &\quad \left. - z^{(1)}(w | \theta) \mathbb{E}[Y_1(w | \theta) \lambda_0\{w c_1(\theta)\} c_1(\theta)] \right\} dw, \end{aligned}$$

with $c_1(\theta) = \exp\{(\theta_0 - \theta)' Z_1\}$ and Conditions C6 and C7 ensuring that θ_0 is unique. Moreover, given that $c_1(\theta_0) = 1$, $\mathcal{U}_{\theta}^{\mathbb{G},\text{eff}}(\delta_0)$ satisfies the condition:

$$\begin{aligned} \mathcal{U}_{\theta}^{\mathbb{G},\text{eff}}(\delta_0) &= \int_0^\infty \left\{ z^{(0)}(w | \theta_0) z^{(1)}(w | \theta_0) - z^{(1)}(w | \theta_0) z^{(0)}(w | \theta_0) \right\} \lambda_0(w) dw \\ &= 0. \end{aligned}$$

Hence, under Conditions C1–C6, the function $\mathcal{U}_{\theta}^{\mathbb{G},\text{eff}}(\delta_0)$ has a non-singular derivative matrix, given by $\Phi_G = \dot{\mathcal{U}}_{\theta}^{\mathbb{G},\text{eff}}(\delta_0)$, which is continuous at $\theta = \theta_0$. Moreover, both $U_{\theta}^{\mathbb{G},\text{eff}}(\delta)$ and $\mathcal{U}_{\theta}^{\mathbb{G},\text{eff}}(\delta)$ exhibit component-wise monotonicity in θ . Finally, let $C_{\epsilon_n} = \{a \in \mathbb{R}^p : \|a\| \leq \epsilon_n\}$ denote a sequence of compact sets, indexed by an arbitrary constant $0 < \epsilon_n < \infty$. To complete the argument, we must show that

the following holds:

$$\sup_{(\boldsymbol{\theta}-\boldsymbol{\theta}_0) \in C_{\epsilon_n}} n^{\frac{1}{2}} \left\{ \frac{n^{-1}U_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta}) - n^{-1}U_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta}_0) - \mathcal{U}_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta})}{1 + n^{\frac{1}{2}} \|\boldsymbol{\theta} - \boldsymbol{\theta}_0\|} \right\} \xrightarrow{P} 0, \quad (\text{A.9})$$

for any positive sequence $\epsilon_n \rightarrow 0$. First, we define $\bar{\delta}_U = n^{-1} \left\{ U_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta}) - U_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta}_0) \right\}$. Then we derive a closed-form expression for $\bar{\delta}_U$ as:

$$\begin{aligned} \bar{\delta}_U &= \int_0^\infty \mathbb{Z}^{(0)}(w | \boldsymbol{\theta}) d\mathbb{N}^{(1)}(w | \boldsymbol{\theta}) - \int_0^\infty \mathbb{Z}^{(1)}(w | \boldsymbol{\theta}) d\mathbb{N}^{(0)}(w | \boldsymbol{\theta}) \\ &\quad - \int_0^\infty \mathbb{Z}^{(0)}(w | \boldsymbol{\theta}_0) d\mathbb{N}^{(1)}(w | \boldsymbol{\theta}_0) + \int_0^\infty \mathbb{Z}^{(1)}(w | \boldsymbol{\theta}_0) d\mathbb{N}^{(0)}(w | \boldsymbol{\theta}_0). \end{aligned}$$

By performing a change of variable in the last two terms of the expansion, applying Lemma A.1, and factoring out the common terms, we obtain:

$$\begin{aligned} \bar{\delta}_U &= n^{-1} \sum_{i=1}^n \int_0^\infty \left\{ \mathbb{Z}^{(0)}(w | \boldsymbol{\theta}) \mathbf{Z}_i - \mathbb{Z}^{(1)}(w | \boldsymbol{\theta}) \right\} Y_i(w | \boldsymbol{\theta}) \\ &\quad \times [\lambda_0 \{w c_i(\boldsymbol{\theta})\} c_i(\boldsymbol{\theta}) - \lambda_0(w)] dw. \end{aligned}$$

Now, we consider a positive sequence of $\epsilon_n \rightarrow 0$, such that $\|\boldsymbol{\theta} - \boldsymbol{\theta}_0\| \leq \epsilon_n$. Then using Condition C1, a Taylor series expansion of $\lambda_0 \{r c_1(\boldsymbol{\theta})\} c_1(\boldsymbol{\theta})$ in the neighborhood of $\boldsymbol{\theta}_0$ yields:

$$\lambda_0 \{r c_1(\boldsymbol{\theta})\} c_1(\boldsymbol{\theta}) = \lambda_0(r) - l_0(r) \mathbf{Z}'_1(\boldsymbol{\theta} - \boldsymbol{\theta}_0) + o_P \left(\|\boldsymbol{\theta} - \boldsymbol{\theta}_0\|^2 \right),$$

where $l_0(r) = \lambda_0(r) + r \dot{\lambda}_0(r) = \psi(r) \lambda_0(r)$. Substituting these into $\mathcal{U}_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta})$, we have:

$$\begin{aligned} \mathcal{U}_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta}) &= -(\boldsymbol{\theta} - \boldsymbol{\theta}_0)' \int_0^\infty \left[\mathbf{z}^{(0)}(w | \boldsymbol{\theta}) \mathbf{z}^{(2)}(w | \boldsymbol{\theta}) - \left\{ \mathbf{z}^{(1)}(w | \boldsymbol{\theta}) \right\}^{\otimes 2} \right] \psi(w) \lambda_0(w) dw \\ &\quad + o_P(\|\boldsymbol{\theta} - \boldsymbol{\theta}_0\|). \end{aligned}$$

Φ_G in (6.1) is obtained by taking the derivative of $\mathcal{U}_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta})$ with respect to $\boldsymbol{\theta}$ in the latter equation. Similarly, we substitute the Taylor series expansion of $\lambda_0 \{r c_1(\boldsymbol{\theta})\} c_1(\boldsymbol{\theta})$ into $\bar{\delta}_U$ to obtain:

$$\begin{aligned} \bar{\delta}_U &= -(\boldsymbol{\theta} - \boldsymbol{\theta}_0)' \int_0^\infty \left[\mathbb{Z}^{(0)}(w | \boldsymbol{\theta}) \mathbb{Z}^{(2)}(w | \boldsymbol{\theta}) - \left\{ \mathbb{Z}^{(1)}(w | \boldsymbol{\theta}) \right\}^{\otimes 2} \right] \psi(w) \lambda_0(w) dw \\ &\quad + o_P(\|\boldsymbol{\theta} - \boldsymbol{\theta}_0\|). \end{aligned}$$

Then subtracting $\mathcal{U}_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta})$ from $\bar{\delta}_U$ results in:

$$\begin{aligned} \bar{\delta}_U - \mathcal{U}_{\boldsymbol{\theta}}^{\mathbb{G},\text{eff}}(\boldsymbol{\delta}) &= -(\boldsymbol{\theta} - \boldsymbol{\theta}_0)' \int_0^\infty \left[\left\{ \mathbb{Z}^{(0)}(w | \boldsymbol{\theta}) \mathbb{Z}^{(2)}(w | \boldsymbol{\theta}) - \mathbf{z}^{(0)}(w | \boldsymbol{\theta}) \mathbf{z}^{(2)}(w | \boldsymbol{\theta}) \right\} \right. \\ &\quad \left. - \left\{ \left\{ \mathbb{Z}^{(1)}(w | \boldsymbol{\theta}) \right\}^{\otimes 2} - \left\{ \mathbf{z}^{(1)}(w | \boldsymbol{\theta}) \right\}^{\otimes 2} \right\} \right] \psi(w) \lambda_0(w) dw. \end{aligned}$$

This implies that:

$$\left\| n^{\frac{1}{2}} \frac{n^{-1} U_{\boldsymbol{\theta}}^{\mathbb{G}, \text{eff}}(\boldsymbol{\delta}) - n^{-1} U_{\boldsymbol{\theta}}^{\mathbb{G}, \text{eff}}(\boldsymbol{\delta}_0) - \mathcal{U}_{\boldsymbol{\theta}}^{\mathbb{G}, \text{eff}}(\boldsymbol{\delta})}{1 + n^{\frac{1}{2}} \|\boldsymbol{\theta} - \boldsymbol{\theta}_0\|} \right\| \leq \frac{n^{\frac{1}{2}} \|\boldsymbol{\theta} - \boldsymbol{\theta}_0\| \|\mathbb{A}\|}{1 + n^{\frac{1}{2}} \|\boldsymbol{\theta} - \boldsymbol{\theta}_0\|}. \quad (\text{A.10})$$

Hence, by Theorem A.2, $\|\mathbb{A}\| \xrightarrow{P} 0$ and given that $\epsilon_n \rightarrow 0$, we obtain (A.11), which ends the proof. $\square$

Proof of Theorem 6.2. The efficient score function for $\boldsymbol{\theta}$ under a general weight function $\mathbb{W}(t \mid \boldsymbol{\theta})$ is given by:

$$U_{\boldsymbol{\theta}}^{\mathbb{W}, \text{eff}}(\boldsymbol{\delta}) = \sum_{i=1}^n \int_0^t \mathbb{W}(w \mid \boldsymbol{\theta}) [\mathbf{Z}_i - \bar{\mathbf{Z}}(w \mid \boldsymbol{\theta})] dN_i(w \mid \boldsymbol{\theta}).$$

First, by applying Theorem A.1 and Corollary A.1, Lemma 1 from [53] implies that there exists a function φ such that:

$$n^{-\frac{1}{2}} U_{\boldsymbol{\theta}}^{\mathbb{W}, \text{eff}}(\boldsymbol{\delta}_0) = n^{-\frac{1}{2}} \sum_{i=1}^n \varphi_i + o_p(1),$$

where $\varphi_i = \int_0^{\bar{\tau}} w(r \mid \boldsymbol{\theta}_0) \{\mathbf{Z}_i - \bar{\mathbf{Z}}(r)\} dM_i(r \mid \boldsymbol{\theta}_0)$, and satisfies the condition:

$$\mathbb{E}[\varphi_1] = \mathbb{E} \left[\int_0^{\bar{\tau}} w(r \mid \boldsymbol{\theta}_0) \{\mathbf{Z}_i - \bar{\mathbf{Z}}(r)\} dM_1(r \mid \boldsymbol{\theta}_0) \right] = 0,$$

by (3.6). Using Lemma 3.1, we have from (3.7):

$$\begin{aligned} \text{Var}(\varphi_1) &= \int_0^{\bar{\tau}} w^2(r \mid \boldsymbol{\theta}_0) \mathbb{E} \left[\left\{ \mathbf{Z}_i - \left\{ \mathbf{z}^{(1)}(r \mid \boldsymbol{\theta}_0) / \mathbf{z}^{(0)}(r \mid \boldsymbol{\theta}_0) \right\} \right\}^{\otimes 2} Y_1(r \mid \boldsymbol{\theta}_0) \right] \lambda_0(r) dr \\ &= \int_0^{\bar{\tau}} w^2(r \mid \boldsymbol{\theta}_0) \left[\frac{\mathbf{z}^{(2)}(r \mid \boldsymbol{\theta}_0)}{\mathbf{z}^{(0)}(r \mid \boldsymbol{\theta}_0)} - \left\{ \frac{\mathbf{z}^{(1)}(r \mid \boldsymbol{\theta}_0)}{\mathbf{z}^{(0)}(r \mid \boldsymbol{\theta}_0)} \right\}^{\otimes 2} \right] \mathbf{z}^{(0)}(r \mid \boldsymbol{\theta}_0) \lambda_0(r) dr, \end{aligned}$$

which shows that $\text{Var}(\varphi_1) = \Sigma_{\boldsymbol{\theta}}^{\mathbb{W}, \text{eff}}$. Consequently, the central limit theorem implies that $n^{-\frac{1}{2}} U_{\boldsymbol{\theta}}^{\mathbb{W}, \text{eff}}(\boldsymbol{\delta}_0) \xrightarrow{d} N(0, \Sigma_{\boldsymbol{\theta}}^{\mathbb{W}, \text{eff}})$, with Condition C9 ensuring that $\|\Sigma_{\boldsymbol{\theta}}^{\mathbb{W}, \text{eff}}\| < \infty$. Under Conditions C1–C9, along with Corollary A.1 and Lemma A.1, the function $n^{-1} U_{\boldsymbol{\theta}}^{\mathbb{W}, \text{eff}}(\boldsymbol{\delta})$ converges uniformly in probability to a function $\mathcal{U}_{\boldsymbol{\theta}}^{\mathbb{W}, \text{eff}}(\boldsymbol{\delta})$, for $\boldsymbol{\theta} \in \mathcal{N}(\boldsymbol{\theta}_0)$. Conditions C6 and C10 guarantee the uniqueness of $\boldsymbol{\theta}_0$. Thus, we define $\Phi_W = \mathcal{U}_{\boldsymbol{\theta}}^{\mathbb{W}, \text{eff}}(\boldsymbol{\delta}_0)$, whose existence is ensured by Conditions A1–A9, while Condition C10 guarantees its nonsingularity. Finally, let $C_{\epsilon_n} = \{a \in \mathbb{R}^p : \|a\| \leq \epsilon_n\}$ represent a sequence of sets, which may not necessarily be compact, indexed by an arbitrary constant $0 < \epsilon_n < \infty$. To conclude the proof, it remains to establish the following:

$$\sup_{(\boldsymbol{\theta} - \boldsymbol{\theta}_0) \in C_{\epsilon_n}} n^{\frac{1}{2}} \left\{ \frac{n^{-1} U_{\boldsymbol{\theta}}^{\mathbb{W}, \text{eff}}(\boldsymbol{\delta}) - n^{-1} U_{\boldsymbol{\theta}}^{\mathbb{W}, \text{eff}}(\boldsymbol{\delta}_0) - \mathcal{U}_{\boldsymbol{\theta}}^{\mathbb{W}, \text{eff}}(\boldsymbol{\delta})}{1 + n^{\frac{1}{2}} \|\boldsymbol{\theta} - \boldsymbol{\theta}_0\|} \right\} \xrightarrow{P} 0, \quad (\text{A.11})$$

for any positive sequence $\epsilon_n \rightarrow 0$. This condition follows from the same reasoning as in the proof of Theorem 6.1. Therefore, applying Theorem 7.3.2 of [54], the results immediately follow. $\square$

Proof of Theorem 6.3. Letting $n \rightarrow \infty$, we obtain the following limits:

$$\lim_{B \rightarrow \infty} \frac{1}{B} \mathfrak{P}' \mathfrak{P} = \begin{bmatrix} 1 & 0' \\ 0 & n^{-1} \hat{\mathbb{A}}_G \end{bmatrix}, \quad \lim_{B \rightarrow \infty} \frac{1}{B} \mathfrak{P}' \mathfrak{Q} = \begin{bmatrix} 0' \\ n^{-1} \hat{\mathbb{A}}_G \Phi_W \end{bmatrix} + o_P(n^{-1}),$$

almost surely, given the data. It follows that $\hat{\Phi}_W = \Phi_W + o_P(1)$, and thus the second result follows from this convergence along with the fact that $\hat{\Sigma}_{\theta}^{\mathbb{W}, \text{eff}} \xrightarrow{P} \Sigma_{\theta}^{\mathbb{W}, \text{eff}}$ under Conditions C1–C10. $\square$

Proof of Theorem 6.4. We can write $\hat{\Lambda}_0(t \mid \theta) - \Lambda_0(t) = \delta_1 + \delta_2$, with $\delta_1 = \left\{ \hat{\Lambda}_0(t \mid \theta) - \hat{\Lambda}_0(t \mid \theta_0) \right\}$ and $\delta_2 = \left\{ \hat{\Lambda}_0(t \mid \theta_0) - \Lambda_0(t) \right\}$. We consider first δ_1 . Under Conditions C1–C10, we can apply Corollary A.1 along with a reasoning similar to that in the proofs of Theorems 6.1 and 6.2 to prove the existence of a function χ such that:

$$\delta_1 = n^{-1} \sum_{i=1}^n \chi_i + o_P\left(n^{-\frac{1}{2}}\right),$$

where $\chi_i = \int_0^t \frac{Y_i(w \mid \theta)}{z^{(0)}(w \mid \theta)} [\lambda_0 \{wc_i(\theta)\} c_i(\theta) - \lambda_0(w)] dw$. Then, assuming that $\|\theta - \theta_0\| \leq \epsilon_n$ for a positive sequence $\epsilon_n = n^{-\frac{1}{2}} \Gamma \rightarrow 0$, we have $\delta_1 = -\Delta'_{\psi}(t) (\theta - \theta_0) + o_P\left(n^{-\frac{1}{2}}\right)$, where $\Delta_{\psi}(t)$ is given in (4.12). This yields the second term of (6.5), showing that $\hat{\Lambda}_0(t \mid \theta)$ is asymptotically linear in θ . Considering now δ_2 , we can use the same Corollary A.1 to show that there exist a function γ , such that:

$$\delta_2 = n^{-1} \sum_{i=1}^n \gamma_i + o_P\left(n^{-\frac{1}{2}}\right),$$

with $\gamma_i = \int_0^t \left\{ z^{(0)}(w \mid \theta_0) \right\}^{-1} dM_i(w \mid \theta_0)$. Substituting $\hat{\theta}_W - \theta_0 = -n^{-1} \Phi_W^{-1} U_{\theta}^{\mathbb{W}, \text{eff}}(\delta_0) + o_P(n^{-\frac{1}{2}})$ into δ_1 and by (6.5), $\hat{\Lambda}_0(t \mid \theta) - \Lambda_0(t)$ converges uniformly in probability to $\bar{\varrho}$, where:

$$\varrho_i(t \mid \theta_0) = \Delta'_{\psi}(t) \Phi_W^{-1} \int_0^{\bar{\tau}} w(r \mid \theta_0) \{Z_i - \bar{Z}(r)\} dM_i(r \mid \theta_0) + \gamma_i$$

Finally, the function $\varrho(\cdot \mid \theta_0)$ is càdlàg and has bounded variation for $t \in [0, \bar{\tau}]$. As a result, it is $\mathbb{P}_0$ -Donsker. Hence, the functional central limit theorem for uniformly bounded empirical processes ensures that the sequence $n^{-\frac{1}{2}} \sum_{i=1}^n \gamma_i$ converges in distribution to a Gaussian process [55]. Its covariance is given by $\mathbb{E} \{ \varrho_1(r \mid \theta_0) \varrho_1(t \mid \theta_0) \}$, for any $r, t \in [0, \tau]$. A direct calculation using Lemma 3.1 and arguments from Proposition 4.2 confirms the covariance expression (6.4); details are omitted for brevity. $\square$

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